MOVEMENT AS EXPOSURE: UNCREWED SYSTEMS AND THE TRANSFORMATION OF LIGHT INFANTRY TACTICS

FROM TACTICAL ADAPTATION TO FORCE DESIGN AND DETERRENCE-BY-DENIAL IN A SMALL STATE CONTEXT

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STUDENT DECLARATION

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Student Declaration

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Abstract

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Contemporary infantry warfare is defined by the transformation of exposure into the central organising variable of combat: who is seen, when they are seen and how quickly observation becomes fire. The literatures on uncrewed systems, military adaptation and mission command rarely connect. Accordingly, drawing on scholarly and doctrinal sources, practitioner interviews and thematic analysis of open-source combat evidence from the Russo–Ukrainian War, this thesis examines how uncrewed systems reshape light-infantry tactics, mission command and small state force design. The democratisation of ISR and precision strike reconceptualises movement as timed exposure across space, time and the electromagnetic spectrum. Persistent surveillance imposes an infantry ceiling, where dispersion preserves survivability but precludes the concentration of mass required for breakthrough. Manoeuvre becomes conditional, achieved through the sequencing of ISR, EW and fires. Tactically, these dynamics drive section-level adaptation: the 14-soldier section emerges as the drone-survival minimum, integrating organic ISR, CUAS, assault pioneering and additional junior leadership. Strategically, for small states such as Ireland, low-cost uncrewed systems offer a means of supporting deterrence-by-denial through disproportionate cost imposition, provided they are integrated into doctrine, training, command, sustainment and force design.

Abstract: Summary of Analytical Contributions

Concept	Analytical Contribution
Movement as Timed Exposure	Reconceptualises movement as the management of exposure across space, time and the electromagnetic spectrum.
Infantry Ceiling	Identifies the threshold at which dispersion precludes the concentration of mass required for breakthrough.
Conditional Manoeuvre	Reframes manoeuvre as a phase enabled by the sequencing of ISR, EW and fires rather than a primary generator of success.
Least Interesting Target	Defines survivability as minimising signature and target value within the enemy's targeting calculus.
FPVD as Organising Weapon	Positions armed drones as an emerging organising weapon around which assault elements may be structured.
Section-Level Adaptation	Demonstrates the requirement for organic ISR, CUAS, IEDD and expanded junior leadership at section level.
Vertical IED Hazard	Extends the IED threat vertically, imposing a render-safe burden and driving the need for organic IEDD.
Mission Command Reinforced	Shows that persistent ISR increases the necessity of decentralised decision-making under degraded control.
UxS and Asymmetric Deterrence	Establishes that low-cost UxS enable disproportionate combat power through cost imposition.
Drone Shock	Identifies the psychological and tactical disruption caused by persistent drone presence, where the sound or expectation of UAS attack alters movement, attention and immediate action.
Relative Combat Power	Demonstrates that UxS impose a disproportionate cost on attacking forces.

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List of Acronyms

AAR After Action Review.

AFRF Armed Forces of the Russian Federation.

AFUA Armed Forces of Ukraine.

ATV All-Terrain Vehicle.

BAI Battlefield Air Interdiction.

BDA Battle Damage Assessment.

C2 Command and Control.

CAS Close Air Support.

CASDs Close Air Support Drones.

CC Critical Capability.

CIED Counter Improvised Explosive Device.

CMO Crisis Management Operations.

CODF Commission on the Defence Forces.

CoG Centre of Gravity.

CR Critical Requirement.

CUAS Counter Uncrewed Aerial Systems.

CV Critical Vulnerability.

DSTL Defence Science and Technology Laboratory.

ECM Electromagnetic Countermeasures.

EDA European Defence Agency.

EMS Electromagnetic Spectrum.

ER Electronic Reconnaissance.

EW Electronic Warfare.

EWSDR Electronic Warfare Software Defined Radio.

FLOT Forward Line of Own Troops.

FPV First-Person View.

FPV-OWADs First-Person View One-Way-Attack Drones.

FPVD First-Person View Drone.

GLOC Ground Lines of Communication.

GWoT Global War on Terror.

HALE High-Altitude Long-Endurance.

IED Improvised Explosive Device.

IEDD Improvised Explosive Device Disposal.

ISR Intelligence, Surveillance and Reconnaissance.

ISRDs Intelligence, Surveillance and Reconnaissance Drones.

LOA Level of Ambition.

LRR Long-Range Reconnaissance.

MALE Medium-Altitude Long-Endurance.

MCADT Marine Corps Attack Drone Team.

MRR Medium-Range Reconnaissance.

NAI Named Area of Interest.

OWAD One-Way Attack Drone.

PPC Phalanx Platoon Concept.

RC Radio Controlled.

RCP Relative Combat Power.

RMA Revolution in Military Affairs.

RUSI Royal United Services Institute.

SLLS Stop Look Listen Smell.

SRR Short-Range Reconnaissance.

sUAS Small Uncrewed Aerial Systems.

TA Thematic Analysis.

TAI Target Area of Interest.

TBI Traumatic Brain Injury.

TTP Tactics, Techniques and Procedures.

UAS Uncrewed Aerial Systems.

UGS Uncrewed Ground Systems.

USMC United States Marine Corps.

UxS Uncrewed Systems — of an unspecified domain.

VOIEDs Victim-Operated IEDs.

YPG Yekîneyên Parastina Gel (AKA People’s Protection Units).

INTRODUCTION

Uncrewed systems have made exposure one of the key problems in land warfare. The literature shows that Uncrewed Systems — of an unspecified domain (UxS) are reshaping reconnaissance, targeting and strike. Wider scholarship on military innovation and mission command explains adaptation under changing conditions. However, these literatures rarely connect. Hence the implications for light-infantry tactics, small-unit command and force design remain underexplored. This thesis addresses that lacuna through qualitative synthesis of scholarly and doctrinal sources, practitioner interviews and systematic analysis of open-source combat evidence from the Russo–Ukrainian War. It identifies recurring patterns in order to derive implications for light infantry adaptation.

Ukraine-focused literature establishes that UxS accelerate Kill-Chains, extend Intelligence, Surveillance and Reconnaissance (ISR) and tighten fires integration, but leaves unresolved how those effects govern light-infantry movement, mission command and small state force design (Cornish, 2024; Molloy, 2024; Bronk, 2025; Watling & Reynolds, 2025b).

At the tactical level, the migration of reconnaissance to drones (Watling & Reynolds, 2025a) has altered the conditions of survivability. Infantry can no longer rely on concealment in proximity to the Forward Line of Own Troops (FLOT). Instead, survivability increasingly depends on dispersion, fortification and Electromagnetic Spectrum (EMS) discipline (Watling & Reynolds, 2023; Watling & Reynolds, 2025a; Watling & Reynolds, 2025b). Since Uncrewed Aerial Systems (UAS) heighten the EMS on the frontline, disciplined signal emission and effective Electronic Warfare (EW) are required, with Electronic Reconnaissance (ER) by overflying systems becoming more common (Watling & Sylvia, 2025). These conditions sharpen the trade-off between communication and detection, placing greater reliance on decentralised execution at lower levels.

However, pervasive ISR simultaneously generates a countervailing tendency. As E. A. Cohen (1996) observed, new technologies enable senior commanders to "perch cybernetically" alongside subordinates. Persistent surveillance intensifies this dynamic, increasing the temptation to centralise control, even as dispersion demands greater autonomy. The literature recognises both trends but does not adequately reconcile them, leaving it unclear how mission command functions in practice under conditions of persistent observation and electromagnetic exposure.

Conceptually, early work anticipated some of these dynamics. Noël (2013: 10) suggested that drones could *occupy* the land from the air. Contemporary evidence indicates that their effect is better understood as a form of aerial denial (Stepanenko, 2025), blurring the concept of the FLOT and constraining manoeuvre and sustainment. Yet the implications of this shift for small-unit tactics remain under-theorised.

The literature on Uncrewed Ground Systems (UGS) further complicates the picture. Rather than mirroring aerial systems, Hinton (2023) argues that their near-term value lies in exposure-reducing support roles — given constraints in mobility, power and teleoperation. As a result, UGS remain comparatively under-explored in both their tactical impact and their implications for light-infantry organisation.

A longstanding sceptical tradition cautions against overstating the transformative impact of new technologies. Krepinevich (1994) argues that invention without organisational adaptation is ineffective, while Betts (1996) highlights the tendency of militaries to misuse new capabilities when doctrine and culture lag behind. More broadly, critics of the Revolution in Military Affairs emphasise evolutionary adaptation over rupture

(Alach, 2008: 50–52), with drone warfare itself often characterised as incremental rather than transformative (Rassler, 2015). Kunertova (2024) extends this critique directly to Ukraine, arguing that observed innovation reflects improvisation under constraint rather than a technological revolution, while Hinote and Ryan (2024) emphasise that drones are only transformative when integrated within effective Command and Control (C2) and sensor networks.

The generalisability of Russo–Ukrainian observations remains contested. Western doctrine, force protection measures and EW capabilities may mitigate some threats, but this assumption has not been tested — particularly at the sub-company level. That leaves the impact of pervasive ISR on section and platoon behaviour unresolved. Ukraine-focused fieldwork documents a shift toward dispersion, fortification and electromagnetic discipline. It does not derive how these principles translate into section Battle Drills. For example, while the ISR mechanism is well documented, its translation into Western doctrine remains underdeveloped.

Pervasive ISR also creates a tension between decentralised execution and the centralising pull of ubiquitous observation. It allows senior commanders to interrogate subordinate action while dispersed units require greater autonomy. Western doctrine emphasises mission command, but is silent on how junior leader initiative operates when multiple echelons observe the same fight. The literature therefore leaves mission command under persistent surveillance insufficiently explained.

Existing studies also treat battlefield effectiveness, organisational adaptation and strategic implications separately. They do not explain what drone warfare means for resource-constrained forces. In particular, they do not show how UxS alter Relative Combat Power (RCP) ratios. This gap is especially acute for small states such as Ireland, whose strategic culture, resource constraints and force structure differ markedly from those of larger powers. Given the operational sensitivity of Tactics, Techniques and Procedures (TTP) evolution, this thesis adopts an indirect analytical approach, examining observable patterns, organisational responses and doctrinal implications at small-unit level through three questions:

1. How have uncrewed systems reshaped the contemporary battlefield, particularly in terms of survivability, sensing, movement and the physical conditions of light infantry tactics?

2. How do persistent ISR and drone-enabled lethality alter light infantry tactics, mission command, section composition and tactical adaptation at sub-company level?

3. What do these tactical and command changes imply for Irish force design, military utility, small state RCP and deterrence-by-denial?

The empirical foundation is a bilateral thematic analysis of approximately 2,000 primary-source items — combat footage, communiqués, imagery and interviews — from the Russo–Ukrainian War. From this corpus, 178 sources were selected, verified and analysed thematically (Braun & Clarke, 2006), coding observable actions into recurring behavioural patterns across survivability, movement, fires coordination, sustainment and command. Examining Ukrainian and Russian material in parallel guards against single-belligerent bias and strengthens pattern validity across the corpus (Webb et al., 1966; Jick, 1979). These sources are treated as partial and success-biased; the analytical question is whether published footage authentically documents the tactical phenomenon under examination, not whether it is statistically representative (Castro, 2013). Secondary sources — peer-reviewed scholarship, practitioner analysis and Irish policy documents — supply conceptual frameworks. Semi-structured interviews with practitioners and subject-matter experts, analysed through an essentialist lens (Finnegan, 2013), contextualise theatre-specific evidence and capture how junior initiative and organisational friction operate in practice. Ethical approval and informed consent governed data collection.

The Russo–Ukrainian War is the principal case because it provides the most demanding observable environment — peer warfare, adaptive and technologically contested. Alternative theatres were considered but offer inferior volume, quality and variety. While Watling (2025a) cautions that its conditions may not translate universally, the underlying mechanisms nevertheless provide analytically transferable lessons. While Ukrainian and Western sources dominate, Russian footage and official commentary were analysed throughout. Open-source video material is subject to selection and success bias and cannot provide statistically representative data. However, the scale, recurrence and cross-source consistency of observed behaviours allow for the identification of tactically significant patterns. This thesis therefore treats such material not as quantitative evidence, but as a qualitative corpus from which recurring adaptations in survivability, fires integration and small-unit behaviour can be derived.

The corpus is deliberately land-focused because the thesis examines light-infantry adaptation. Its main limitation is that the Russo–Ukrainian War is ongoing and many relevant lessons are classified, or not yet codified in doctrine, capability codes or capability statements. Hence, institutional outputs lag battlefield practice. The author's position as a serving Irish Army officer also shapes the analytical frame, particularly the emphasis on section-level adaptation, mission command and small state force design. The thesis therefore draws conclusions from observable tactical behaviour, practitioner evidence and open-source documentation. It uses bilateral Ukrainian/Russian coding, Cyrillic primary sources and triangulation to identify recurring mechanisms. This thesis is intended to be actively disseminated among infantry units, junior commanders and Defence Forces' staff officers concerned with the implications of persistent ISR for light infantry, force design and deterrence-by-denial.

Chapter one shows how UxS have democratised ISR and precision strike, making exposure a core tactical problem. It establishes the battlefield conditions on which the rest of the thesis builds. Chapter two shows how persistent surveillance changes light-infantry tactics, command and organisation. It develops the concepts of movement as timed exposure, the infantry ceiling and conditional manoeuvre, before proposing adapted section structures built around organic ISR, Counter Uncrewed Aerial Systems (CUAS), Assault IEDD/Assault Pioneering and stronger junior leadership. Chapter three examines how drone-enabled mass and cost asymmetry reshape force design and deterrence-by-denial. It argues that small states can generate disproportionate RCP through integrated UxS, enabling credible deterrence-by-denial under resource constraints.

1 WHAT HAS THE BATTLEFIELD BECOME?

UxS have made movement one of the most dangerous actions on the battlefield. In the *grey zone*, exposure to observation dictates risk. This chapter examines how democratised surveillance and armed UxS have reshaped warfare.

While context-dependent, the Russo–Ukrainian War exposes lessons which are applicable to militaries worldwide. Molloy (2024: 9–13) and Zabrodskyi et al. (2022: 62–63) describe the adoption of drones since the outbreak of hostilities. The war's positional (Watling, 2024), attritional (Zabrodskyi et al., 2022; G. S. Jones et al., 2023; Watling & Reynolds, 2023; Watling & Danylyuk, 2025; Watling & Reynolds, 2025a; Watling & Reynolds, 2025b) character is the product of mutually-denied airspace operating under the shadow of nuclear deterrence, rather than manoeuvre's obsolescence (Bronk, 2022: 5:20–10:20), (Watling, 2022: 25:00–27:00) and (Hvizda et al., 2025: 22–23, 31).

However, Western militaries must resist uncritical lesson importation. Watling (2023b: 1–2) argues that abstract demonstrations of capability often obscure the enabling requirements necessary to make them plausible in practice. Some analysts consider that drones have rendered manned airpower irrelevant (Barno & Bensahel, 2024), while others caution that armoured and air warfare remain crucial (Shaikh & Rumbaugh, 2020; Barrie & Ebert, 2021; Bronk, 2025). Interview Participant 08 (2026) assessed that combined-arms manoeuvre could exploit the positional dependence of drone teams, dislocating their positions and reducing their battlefield effect. Cornish (2024: 5:00–6:00) situates drone warfare within a broader taxonomy of ideational, kinetic, technological and cognitive warfare, cautioning against overemphasising the technological dimension — echoing earlier scholars' perspectives. The offence–defence balance currently favours the defender[1].

Zaluzhny's (2026) contention of Ukraine as a "laboratory of new tech" warrants examination. Militaries are often accused of preparing to fight the last war, yet there is strategic value in seeking to close peacetime gaps in RCP. States cannot know in advance whether war will merely expose peacetime weaknesses or punish them. The safest peacetime strategy is to seek parity — underscoring the relevance of the ongoing war.

Watling (2025a: 15) considers pervasive ISR to be "the single most transformative element of warfare in Ukraine". Interview Participant 10 (2026) cautions against dismissing UxS as a fleeting phenomenon and against overcorrecting through radical force redesign, noting that comparatively low cost equipment and TTPs may suffice. Meanwhile Interview Participant 08 (2026) cautions against bureaucratic inertia, citing Paine's (1776) "a long habit of not thinking a thing wrong gives it a superficial appearance of being right".

Weather and terrain create windows for action which light infantry must learn to exploit. Radio Controlled (RC) UAS usually operate within the microwave band which is attenuated by liquid water (Vollmer, 2025). Hence wet environments, rain and humidity reduce their range. Greater battlespace transparency heightens the importance of concealment. While poor weather has long enabled concealed movement, modern tactics place a premium on conditions inhibiting small rotary-wing UAS (Civ Division, 2025d: 3:15–4:56). Fog obscures visual detection, while headwinds degrade endurance (Civ Division, 2026d: 0:10–0:56) — exacerbated by their circa 25-minute lithium-polymer battery life (Civ Division, 2026c; Interview Participant 01, 2026). Water vapour degrades uncooled thermal imaging. Consequently, UAS employed around the grey zone remain highly weather-

[1] Though both sides continue to adapt; see G. S. Jones et al. (2023) for battlefield evolution.

dependent (Kunertova, 2024; Miller et al., 2026: 10). Preferential times for action include poor weather, dusk and dawn — reminiscent of ISIS' *thunderstorm assaults*.

Broken terrain and clutter — which is separated on the order of the signal's wavelength — limits line of sight, reduces the range of RC drones (Néron-Bancel & Garnier, 2024: 31,53) and is compounded by the opacity inherent in the land environment. This can be mitigated by employing UAS to rebroadcast signals. Since RC range is highly sensitive to terrain, First-Person View One-Way-Attack Drones (FPV-OWADs) remain vulnerable even with relay support: after reaching the target area, they may be unable to descend to attack altitude without dropping the control link. Close Air Support Drones (CASDs) with higher operating altitudes and less stringent line-of-sight requirements, are somewhat less exposed to this constraint. Hence, close terrain reduces the persistence and pervasiveness of RC UAS (Civ Division, 2026b: 1:00–1:57) and RC UGS. Furthermore, for similar reasons, dense terrain reduces the range of EW jamming. Unlike the flat nature of Ukrainian steppe where EW transmitter range is optimised, close terrain will necessitate careful planning and judicious selection of target areas of interest.

While ISR is increasingly effective on the modern battlefield, the trend exemplified in Ukraine can be misleading if generalised (Interview Participant 10, 2026). The flat steppe and limited vegetation are exceptionally transparent to ISR collection, maximising detection range and hindering safe movement. Closer and hillier terrain provide opportunities to remain undetected by channelling surveillance to named areas of interest. For other areas, concealment may once again offer meaningful mitigation against both observation and engagement, reducing the near-equivalence seen in Ukraine.

Assertions that modern technologies have eliminated the *fog of war* and produced a transparent battlespace are not new. They have consistently proven overly optimistic

(Alach, 2008: 49). E. Davis (2025) cautions against concluding that battlefield transparency will ever be fully realised. Interview Participant 10 (2026) argues that Ukraine's open terrain offers near-ideal conditions for ISR drones, making it misleading to attribute transparency solely to drones rather than to the terrain they exploit.

Human factors constrain drone observation. Pilots perceive the battlespace through a two-dimensional visual interface, lacking depth perception and peripheral awareness. This limits their ability to interpret texture and subtle environmental changes — emphasising movement as the dominant detection cue. The Stop Look Listen Smell (SLLS) drill provides contextual understanding which remote sensing cannot replicate (Civ Division, 2025d: 4:35–5:32).

As battlefield observation becomes cheaper and more widespread, close-air lethality increases and exposure becomes more costly (Hvizda et al., 2025: 15) and (Néron-Bancel & Garnier, 2024). State Border Guard Service (2023) depicts how light forces without CUAS are vulnerable to attrition by ISR-corrected fires before they can close with the enemy. Kraken (2022) demonstrates that Ukraine's early-war tactics followed well-understood Western combined-arms doctrine. These methods are no longer survivable. Western forces have shown marginal and uneven adaptation. The French Foreign Legion's *Gévaudan 26* exercise is illustrative: despite reported First-Person View Drone (FPVD) integration (Ministère des Armées, 2026), the observed use of tight formations, lack of CUAS sentries, exposed movement and force concentration remains inconsistent with drone warfare (TF1 INFO, 2026). NATO's Exercise Hedgehog further demonstrates the disproportionate lethality which skilled UAS operators can generate (Melchior, 2020).

Interview Participant 04 (2026) noted that in 2022, infantry could still "scurry into a wood line" and wait for surveillance drones to leave; they "didn't have persistence". By

2023–24, as Interview Participant 09 (2026) describes, the conflict had shifted: both sides were employing layered Intelligence, Surveillance and Reconnaissance Drones (ISRDs), while loitering munitions and FPV-OWADs converted detection into almost immediate lethality (Zabrodskyi et al., 2022; G. S. Jones et al., 2023; Watling & Danylyuk, 2024; Watling & Reynolds, 2025a; Interview Participant 10, 2026). Ukrainian tactics have evolved under the experience of frontline conditions often involving up to 25 UAS tracking a single formation (Watling & Reynolds, 2023: 32). Sections can expect four ISRDs to be simultaneously overhead — often with friendly drones indistinguishable from the enemy's — creating fratricidal and other operational frictions (Interview Participant 09, 2026). Both belligerents field ISRDs at platoon level, with commensurate EW countermeasures on each side (Hvizda et al., 2025: 14).

This logic collectively embodies what this thesis terms *movement as timed exposure* — routes must be pre-planned and remotely reconnoitred; soldiers move quickly in dispersed packets between cover and concealment; pauses are required to check for drones; and delay increases the likelihood of detection and engagement.

Reports such as Molloy (2024: 22) and veterans such as Interview Participant 01 (2026) describe the grey zone or *zone of contestation*, where mutual sensor-fires parity makes concentration prohibitively dangerous. This view enjoys broad scholarly support: Néron-Bancel and Garnier (2024: 23), Tollast (2025), Watling (2023a: 17:21–22:37), Watling (2023b: 53–58), Watling (2025a: 2–3) and Zabrodskyi et al. (2022: 53–55). Extending 15–30 kilometres past the FLOT — or *zero line* — (Watling, 2025a; Tyshchenko & Kiley, 2026: 5), the grey zone has supplanted the traditional *close–deep–rear* model with zones of *opportunity–contestation–risk* (Watling, 2023b: 53–58). Shaping operations must now position objectives within the friendly zone of opportunity prior to their assault.

Cheap UAS have delivered persistence and rapidly scalable strike at the small unit level (Civ Division, 2026d: 4:11–5:12), (Kunertova, 2024: 23), (Néron-Bancel & Garnier, 2024: 26–28) and (Watling, 2025a: 2). Tollast (2025) emphasises that ISRDs at lower echelons are central to transparency. Their utility in light infantry assaults is striking: they enable real-time adjusting of fires, battle damage assessment and decision-making.

The mass arrival of Russian FPV-OWADs "completely turn[ed] the tide" (Interview Participant 04, 2026; Interview Participant 09, 2026). Detection–to–engagement times have fallen to as little as three seconds (Interview Participant 08, 2026). ISRDs also compress the kill-chain, providing target coordinates directly to fire controllers (Molloy, 2024; Néron-Bancel & Garnier, 2024: 29). Movement corridors and logistics routes become "predictable and targetable", degrading tempo and increasing attrition (Watling & Danylyuk, 2024).

On the ground, this persistence is experiential: "the buzz of rotors overhead is constant", with fibre-optic cables and netted roads physically marking a battlespace (Miller et al., 2026). For example, the 225th Separate Assault Regiment (2022) depicts the first-person view of a motorised infantry mortar team being effectively targeted by FPVDs. Veterans now describe the Russo–Ukrainian battlespace as: "impossible not to be seen" (Interview Participant 01, 2026), "you will get seen, you will get attacked" (Anonymous Veteran of the Russo-Ukrainian War, 2026).

Infantry have adapted through dispersed, "packeted" infiltration, deliberate SLLS *drone halts* and strict signature discipline (Zabrodskyi et al., 2022; Interview Participant 01, 2026; Interview Participant 06, 2026; Interview Participant 09, 2026) and (Civ Division, 2026d: 8:11–8:23). Command posts and drone teams likewise minimise electromagnetic emissions and disperse to survive fires (Watling, 2023b). Dispersion disaggregates the force: small elements operating under signature discipline cannot be centrally controlled. Indeed, Watling and Reynolds (2025b: 9) report section defences with as much as 200 m frontage. This places a premium on subordinate initiative and mission command.

Greater battlefield transparency imposes a structural trade-off between survivability and the ability to generate effects (Watling, 2023a: 4:36–6:06). Survivability can be enhanced through concealment or hardened positions, but these constrain effective action. Subterranean positions illustrate this: their entrances must be secured against drones and infantry, constraining manoeuvre (Civ Division, 2026f). Interview Participant 04 (2026) describes assaulting positions where entrances had been welded shut — prioritising protection at the cost of tactical flexibility — and apparently armed with an "unlimited amount of grenades".

Armed UAS have produced an asymmetric cost imposition which fundamentally changes the economics of attrition for light forces. Lethality in the Russo–Ukrainian War has exhibited a marked switch from artillery to drones (Zafra et al., 2024), where a purported 80–90 percent of casualties are caused by UAS (Crawford, 2026; Hartog, 2026; Zelenskyy, 2026) — which is consistent with the expectations of this research for light forces. This reflects a substantial asymmetry in cost and effect. Magyar (2026a) reports that *Magyar's Birds* and associated UAS units account for over a third of verified Russian losses while comprising just two percent of the army's headcount, at a reported cost of $878 per enemy soldier killed — a striking asymmetric cost. Their comparatively low cost reduces the penalty for inefficiency, enabling redundant expenditure of drones on a single target. 10" CASDs and 5" FPV-OWADs can be expected to deliver light-mortar sized warheads at ranges of seven kilometres (3.5 return) for circa $500 (Civ Division, 2025c: 2:08–3:20). While radio-controlled systems remain the most common (Civ Division, 2026e: 0:04–0:18), fibre-optic tethering provides resilience against jamming and extends operational range to tens of kilometres. Molloy (2024: 10) identifies the prevalence of small drones as the most frequently cited theme among participants. As these systems operate predominantly at the tactical level alongside infantry, this reinforces the need to integrate UxS into the planning and preparation of infantry forces.

While Watling (2025a: 15) acknowledges the "utility" of FPV-OWADs, two of his suggested limitations merit closer scrutiny. First, the ease of protecting positions against them appears overstated for light infantry: positions are small (vulnerable to blast) and require firing ports, creating unavoidable vulnerabilities which FPVDs can exploit. Second, their purported vulnerability to interception underestimates the effects of their small size, high speed and frequent nocturnal employment (constituting a particular light infantry vulnerability). Leppänen's (2025) field demonstration reinforces the central claim of this thesis: that light infantry positions are fundamentally exposed to FPV-OWADs under realistic conditions. In a fashion analogous to human ambushes, Waiter Drones (waiters) attempt to optimise concealment with observation of a target area — possibly for weeks (Stepanenko, 2025). While they are vulnerable to being pre-seen and destroyed — particularly where the drone has a single camera — detection is not usually a warning but rather an immediate prelude to strike.

Dropper drones — typified by systems such as AFUA's *Baba Yaga* — represent the most functionally versatile UxS in the contemporary battlespace. Unlike FPV-OWADs, they are reusable, payload-flexible and multi-role: delivering CAS, acting as motherships, emplacing mines along logistics corridors and conducting resupply (Azov Media, 2025a; Watling, 2025a; Anonymous Veteran of the Russo-Ukrainian War, 2026: 8). Watling (2025a: 12) assesses CASDs as among the deadliest tools in Ukraine's arsenal, with innovations including double-drops and magazine-fed release systems increasingly emulating crewed aircraft (Murray & Tuga, 2025; Taylor, 2026). Their significance extends beyond lethality. By assuming the most exposure-intensive tasks — last-mile resupply and route interdiction — dropper drones displace humans along predictable, targetable routes. This directly mitigates the hazards of human movement. While scholars emphasise the enduring primacy of indirect fires on grounds of responsiveness, payload and range, CASDs contest each criterion: they deliver repeated precision strikes with substantial explosive payloads and operate from positions already near the FLOT — compressing effective range without the signature or positional constraints of tube systems.

UGS do not replace infantry; they displace the most dangerous human movement, thereby sustaining combat power under persistent observation. The Russo–Ukrainian War has accelerated the adaptation of UGS for logistics, reconnaissance and fire support — particularly since the failure of Ukraine's 2023 offensive (Kofman & Lee, 2023) and (Watling, 2025a: 4–5). Dedicated units are emerging which employ UGS for assault, mine-laying, resupply and CASEVAC, offering light infantry a means to sustain combat effectiveness while mitigating exposure to increasingly lethal and transparent battlefields (Bendett, 2024; Kirichenko, 2025; Third Army Corps, 2025b; Watling, 2025a). Where personnel are constrained, drones allow fewer soldiers to be exposed to risk while extracting greater utility from limited human resources over time. Watling's (2023b: ch. 9) conclusions regarding the future offensive and logistical use of UGS by light infantry have largely been realised.

UGS differ fundamentally from UAS since they must negotiate the physical complexity of the terrain rather than overfly it. Constrained by undulations, debris, gradient and soil conditions, tipping or immobilisation is a recurring vulnerability. Although their low profile may reduce detection and enhance survivability (Interview Participant 02, 2026), ground navigation imposes difficulties absent from aerial systems. This is consistent with the “hidden difficulty” described by Rivero (2026), reported by others (Manley, 2024) and experienced by the author.

The implication is that effective UGS employment is more terrain- and weather-dependent than aerial systems. This demands a level of operator judgement and local adaptation which cannot be scripted from above, reinforcing the premium on decentralised control examined in Chapter 2.

UGS sustain humans under conditions in which movement itself has become a primary vulnerability. As Molloy (2024: 33) identifies, drones displace rather than replace humans. It is this displacement which sustains combat power under persistent observation — a logic now operationalised at scale in Ukraine — as illustrated by Russian Ministry of Defence (2026a). There, UGS assume the final, most dangerous leg of sustainment after crews deliver them by vehicle to a forward release point, beyond which human drivers and stretcher-bearers can no longer operate safely (Interview Participant 02, 2026).

At Pokrovsk — a heavily contested Russo–Ukrainian War axis — the displacement was already extensive. Shatalov (2025) (a frontline commander), Third Army Corps (2025b), (Interview Participant 02, 2026) and Rivero (2026) report that approximately 80–90 percent of forward resupply and casualty evacuation are conducted by UGS. This reflects what Watling (2023b: 57) identifies as the core utility of robotics for dispersed light infantry — sustaining them without requiring the movement that would expose soldiers to detection and targeting. It should be noted that CASDs are frequently re-rolled as logistics drones (Watling, 2025a: 18).

The operational stakes were exemplified near Kostiantynivka. UGS delivered medical supplies, a stretcher and warm clothing under simultaneous artillery and FPV attack, with a second vehicle completing the evacuation of wounded soldiers after repeated human attempts had failed or generated further casualties (Bendett, 2024; Kirichenko, 2026). UGS therefore do more than reduce risk. In a battlespace where grey zone approaches are saturated with ISRDs (Interview Participant 02, 2026), UGS make "last mile resupply" and casualty recovery possible where manned movement has become operationally infeasible (Lyle, 2023a: 40:00–41:52). Indeed, Taylor (2026: 18:50–28:00) reports UGS range greater than 20 kilometres.

The displacement of human movement by machines represents a structural shift in how combat power is sustained. By transferring risk from personnel to platforms, robotics mitigate the vulnerability of light infantry to bypass, positional warfare and encirclement. This logic extends beyond peer warfighting, offering a means to reduce the *targetable action* inherent in resupplying forward operating bases and isolated positions in inter-positional operations such as crisis management and counter-insurgency.

Interview-based reporting from a Ukrainian UGS company commander indicates that armed UGS are used for support-by-fire, harassment, breaching defences, sabotage, ambushes and reinforcing infantry assaults at the tactical level (Suprun, 2025; Third Army Corps, 2025c; DevDroid, 2026). Ukrainian payloads from tens to hundreds of kilograms are reported possible (UNITED24 Media, 2024). UGS can deliver substantial explosive payloads for breaching defences and destroying known positions, illustrated by Fedorov (2024) and Mingus and Harris (2026).

The complexity and difficulty of piloting and navigation likely limit the utility of UGS during fast-paced manoeuvre. However, their value in breaching, suppression and attrition is evident, echoing earlier uses of uncrewed systems by ISIS (Tveterås, 2022) and bearing comparison to the static logic of First World War mine warfare. Their potential as disaggregated support-by-fire platforms is likewise significant, as illustrated by Watling (2025a: fig. 4). This suggests that the tripod-mounted machine-gun as the primary light infantry direct fire weapon may face structural redundancy in this role. The skilful use of ISRD-corrected machine guns in the *indirect-fire* role has also been observed. UGS are suited to urban operations, where they can conduct reconnaissance by fire and provide heavy direct-fires, while mitigating human exposure in these challenging roles (Watling, 2023b; Interview Participant 08, 2026: 151). They are also useful for reconnaissance during building clearance, although extensive rubble may favour UAS. Future quadrupedal robots may offer a replacement for pack animals in difficult terrain, enhancing light infantry sustainment while increasing combat effectiveness and lethality. This reflects a structural shift in which firepower can be projected without exposing personnel, with operators controlling systems from protected positions or armoured vehicles.

In July 2025, Ukraine's 3rd Separate Assault Brigade reportedly induced the surrender of Russian troops using FPV-OWADs and UGS, suggesting that UxS may influence not only physical survivability but also an adversary's will to fight (Euromaidan Press, 2025; Zoria, 2025; Mingus & Harris, 2026). The psychological effects of human–machine combat remain underexplored.

Armed UGS provide a means to project defensive firepower while reducing human exposure, functioning as platforms for infantry support (Interview Participant 02, 2026). Ukrainian experience demonstrates their growing institutionalisation, with specialised units — such as the NC13[2] — employing robotic systems for support, ambush and defen-

[2] A strike UGS company of the 3rd Assault Brigade.

sive tasks. It reflects a broader effort to replace personnel in the most dangerous sectors of the front (Suprun, 2025; DevDroid, 2026).

In practice, armed UGS have demonstrated the ability to assume limited defensive roles, including replacing infantry positions and maintaining continuous fire over key terrain for extended periods, thereby relieving pressure on manpower-constrained units (Shumlianskyi, 2025). However, their utility remains constrained by vulnerability to UAS, limited endurance, communication fragility and the requirement for human support to sustain operations, maintain UGS and ultimately occupy terrain (Interview Participant 02, 2026). Consequently, while they enhance defensive capacity and redistribute risk, they do not replace infantry and may be better suited to covering gaps along quieter sectors of the front — providing nuance to recent rhetorical claims (Third Army Corps, 2025a; Loh, 2026). Moreover, their relative cost compared to aerial drones imposes compromise: while they reduce risk to personnel, they are not readily expendable and must be employed selectively within a broader human–machine defensive system. While CASEVAC UGS and dispersed infantry seek to survive by reducing their signature and becoming less attractive targets, logistics and defensive UGS may instead constitute *more interesting* high-payoff targets for the adversary (Interview Participant 02, 2026) and (Lyle, 2023a: 40:00–42:00).

Infantry have adapted by carefully planning their movements, using dispersion, signature discipline and deliberate pauses to survive. Practitioner evidence suggests that drone avoidance requires a fundamental shift regarding how light infantry manage exposure, time and signature. As an Anonymous Veteran of the Russo-Ukrainian War (2026) emphasises, traditional camouflage principles remain central. UAS detection is primarily movement-based; identification via shape, shine, shadow, spacing or silhouette is far less reliable without AI (Civ Division, 2026c: 5:00–6:00). However, camouflage is best considered a component of one's survivability equipment (Hall et al., 2021) — comparable to body armour. The democratisation of precision strike described by Interview Participant 08 (2026) and Hvizda et al. (2025: 14) results in camouflage shaping whether a soldier is detected, which is the first stage in the adversary's kill-chain (Hémez, 2017). Watling (2025a: 3) observes Russian forces employing thermal sheeting. This is an adaptation not merely to visual detection but to multi-spectral surveillance — a shift further illustrated by UA Navy (2025) and UNITED24 Media (2026).

Crucially, the tempo of the drone-enabled kill-chain forces a shift away from static concealment towards dynamic survivability. This often requires minimising direct exposure to the sky, though doing so may create tensions with dispersion when concealment is limited. While earlier Russian practice — taken from blogger Filatov — emphasised freezing to avoid detection, emerging evidence suggests this is increasingly untenable against modern FPV systems. Combat footage reviewed for this research indicates that find–engage cycles are measured in seconds. For planning purposes, sub-minute engagement timelines will outpace human reaction. Hence hesitation or indecision can be fatal. This is reflected by Filatov (2025a), who highlights that improvements in sensors and operator proficiency have improved the detection of stationary targets. Consequently, dispersion, rapid movement and the exploitation of terrain are essential immediate actions for survival (Civ Division, 2025b).

This creates a fundamental tension in infantry behaviour: movement increases the risk of detection, yet immobility increases the risk of engagement once detected. Commanders must continuously balance exposure against survivability, making rapid decisions on whether to halt or manoeuvre based on incomplete information. The emphasis on SLLS halts reflects an attempt to avoid detection and the dilemmas arising if observed — producing movement akin to ‘island hopping’.

The variation in drone types further complicates this tactical problem. The persistence and altitude of fixed-wing ISR systems mean that infantry must often assume continuous observation. The shorter range of rotary systems creates opportunities to exploit their poor endurance and even to locate and target drone operators (M. Jones, 2025: 6:52–13:45). Indeed, waiter drones have been used to triangulate FPVD teams by watching other drones’ flight paths. This suggests that drone avoidance is not purely defensive: in certain conditions, it enables a reversion to traditional infantry strengths, including stalking and counter-reconnaissance. Survival is therefore not about avoiding drones altogether. It is about reducing detectability long enough to avoid being pulled into the kill-chain (Néron-Bancel & Garnier, 2024: 29).

Drones make routine movement along previously ‘proven routes’ increasingly hazardous. For example, waiters may ambush such routes, while CASDs may drop landmines and IEDs. Hence, the concept of long-term proven routes becomes increasingly obsolete. Ukrainian SOF practice has institutionalised this logic through structured pre-movement planning, integrating UAS reconnaissance, UAS overwatch, dedicated CUAS groupings and continuous route reassessment as baseline requirements (Ukrainian Lessons Learned SOF department, n.d.-a; Ukrainian Lessons Learned SOF department, n.d.-b). Route selection and adaptation cannot reside with higher command; the tempo, decentralised nature and unpredictability of drone-enabled interdiction push these decisions downward to the lowest level. Given individual ownership of risk during movement, some Ukrainian units performing infiltration are reported to adopt a decentralised, collaborative planning approach in which the full platoon participates in the estimate rather than receiving orders from above (Leach, 2025: 11:47–15:00) — a dynamic examined in Chapter 2.

Under pervasive ISR, the protective qualities of speed and dispersion are emphasised (Zabrodskyi et al., 2022: 62–63). Néron-Bancel and Garnier (2024) and Mingus et al. (2026: 65) reduce this logic to a blunt rule: "mobility equals survivability" in a manner at odds with recent Western approaches to protection. The practical expression of this is visible in the proliferation of light mobility platforms — such as motorcycles and all-terrain vehicles — appearing on the battlefield. Yet speed alone does not confer immunity. For example, Filatov's (2025b) footage captures the precarious nature of grey zone movement, narrowly avoiding an FPVD strike even when moving at speed. Mobility raises the cost and difficulty of targeting but the significant risk to infantry remains. It creates tensions with the logic of being uninteresting, as the speed and signature required for movement may themselves attract attention. The margin between survival and attrition is often measured in reaction time rather than distance — especially since vehicles cannot outrun FPVDs. Hence, movement must be aggressive and decisive. However, speed introduces its own hazards: high-tempo movement across broken terrain frequently results in crashes, loss of control and personnel being thrown from vehicles. Speed is therefore not a safeguard, but a trade-off, reducing exposure to the enemy kill-chain.

Thermal sensing further complicates drone avoidance by challenging assumptions about concealment. In terrain permissive to ISR, concealment at night can be more difficult than during daylight, as reduced ambient heating and background clutter increase the detectability of human heat signatures (Civ Division, 2025b: 2:15–3:00). This inverts the traditional assumption that night movement is inherently safer, with thermal sensors in some cases rendering nocturnal manoeuvre more vulnerable than daylight movement.

Even attempts at thermal camouflage may prove counterproductive, as coverings can generate distinct silhouettes or re-radiate absorbed heat following contact. Under such conditions, daylight movement may present lower detection risk, particularly where visual concealment remains effective and detection relies more heavily on movement than static signature (Civ Division, 2025d: 8:14–8:20). While low-cost mitigation measures such as scrim netting and improvised overhead cover do not eliminate thermal vulnerability, they can introduce friction into the detection process, delaying the transition from observation to engagement (Civ Division, 2025b: 0:16–50).

A functional divergence has emerged between endurance-oriented defenders in dispersed hardened positions and highly-trained assault elements conducting small-team infiltration. This reflects adaptation rather than doctrinal preference. Watling and Reynolds (2025a: 8) note a wide variety of ability amongst Russian infantry; poorly trained soldiers are used for infiltration and reconnaissance by fire while experienced and more capable troops are preserved for assaulting.

UxS have fundamentally altered the conditions under which light infantry fight, survive and move. The battlespace is no longer opaque; movement has become timed exposure; the grey zone has replaced the traditional close–deep–rear model with overlapping zones of opportunity, contestation and risk. UGS extend this logic into sustainment, displacing human movement from the most lethal ground and enabling persistence where exposure would otherwise be prohibitive. Yet these adaptations bring their own constraints: dispersion preserves short-term survival but erodes mass and tempo. Signature discipline and fastidious route planning become Battle Drills. The result is a trade-off between survivability and combat power. Light infantry remain indispensable for clearing and consolidating ground, but now function primarily as survivable shaping and fixing elements within a wider human–machine system that generates manoeuvre through the sequencing of effects across forces and domains (Watling, 2025a: 5–11). The next chapter examines how these conditions reshape mission command and force design at the lowest tactical level.

2 HOW INFANTRY MUST FIGHT AND COMMAND

Rather than rendering mission command obsolete, the Russo–Ukrainian War has reinforced its necessity. Democratised ISR and precision fires expose C2 to systematic detection and strike — driving dispersion which prioritises subordinate initiative. As Busby (2025: 208) identifies, mission command's core function is to reduce battlefield uncertainty under such conditions. What changed was command post survivability. Drones make legacy headquarters systematically vulnerable — Chornobaivka demonstrated tactical echelons could be found and struck (Beagle et al., 2023). Mission command becomes a necessary mitigation as HQs become smaller, dispersed and less visible.

Metz (2000) argued that doctrines such as AirLand Battle blended modern technology with mission command and tempo, indicating that technology and decentralisation are not inherently incompatible. Digital systems create persistent incentives toward oversight, even when doctrine prescribes restraint. Krulak's (1999) "strategic corporal" reflected the link between tactical action and strategic consequence. The legal imperatives of the Global War on Terror (GWoT) further institutionalised this dynamic, drawing senior commanders into increasingly granular oversight. As Betts (1996) observes, technology does not improve judgement; it amplifies existing organisational tendencies. Such tendencies are evident among AFRF and AFUA formations, where distant commanders dictate individual soldiers' movements and actions (Interview Participant 01, 2026). ISRDs therefore have an inherent tension by simultaneously enabling centralised control and subordinate initiative. Contemporary mission command is thus less a settled doctrine than a contingent practice: the virtual oversight described by E. A. Cohen (1996) persists, but the dispersion required for survival reinforces the decentralised execution anticipated by United States Marine Corps (2016: 15).

Persistent ISR and drone-enabled lethality have forced light infantry to fundamentally rethink tactics, mission command and unit composition at sub-company level. Battlefield effects are increasingly generated prior to contact — by machines — through integrated reconnaissance, strike and counter-reconnaissance systems. Hence, infantry must assume their movements are constantly observed and plan to employ and be engaged by machines, as advised by US Department of the Army (2025: 58–63).

While debate persists over the utility of larger drones (Calcara et al., 2022), there is little uncertainty regarding the effectiveness of small drone technology's impact on light infantry. Considering Lanchester's force equations — where Equation 2.1 depicts the relative loss rates[3] (Lanchester, 1916; Epstein, 1988; Mearsheimer, 1989; P. K. Davis, 1995; Hughes, 1995) — drones do not make the Russo–Ukrainian War reducible to a pure Lanchesterian duel, in which attrition is treated as mathematically predictable from force strength and lethality coefficients alone. Analysing the lethality coefficients is incomplete, because they are shaped by EW, weather, adaptation and force heterogeneity. Nevertheless, they make local breakthrough combat more Lanchesterian in character by raising the salience of relative lethality and local exposure. Even where both sides possess surveillance and strike drones, the attacker must mass, move, breach and remain observable for longer. The defender by contrast remains comparatively concealed in prepared positions. Battlefield transparency therefore affects attrition asymmetrically. It does not merely improve observation in general. Instead it raises the defender's effective lethal-

[3] Where RLR_i is the ratio of attacker to defender fractional loss rates on sector i, $K_{d,i}$ and $K_{a,i}$ are defender and attacker lethality coefficients, and $F_i = A_i/D_i$ is the local force ratio.

ity coefficient $K_{d,i}$ relative to the attacker's, by increasing the probability of detection, effective engagement and lethality.

$$RLR_i \equiv \frac{ALR_i}{DLR_i} = \frac{K_{d,i}}{K_{a,i}} \frac{1}{F_i^2} \tag{2.1}$$

$$F_{i,\text{breakeven}} = \sqrt{\frac{K_{d,i}}{K_{a,i}}} \tag{2.2}$$

Setting $RLR_i = 1$ gives Equation 2.2's local break-even ratio, of which the 3:1 rule is simply the case $K_{d,i}/K_{a,i} = 9$ (Mearsheimer, 1989). Any UxS-driven gain in the defender's relative lethality forces the attacker's required ratio up non-linearly. Numerical mass becomes a progressively poorer substitute for positional advantage. The result is that the historical 3:1 heuristic likely understates the local superiority now required for successful breakthrough in drone-saturated positional warfare. As Watling (2023a: 40:33–44:00) notes, drones also narrow the technological gap between peer and sub-peer forces.

Without reliable CUAS (including EW and manoeuvre) UAS become a primary driver of light infantry attrition: directly as OWADs and CASDs or indirectly through ISRD-corrected fires. Molloy (2024: 29) argues that this is particularly consequential for light infantry operating without reliable access to suppressive fires (Kunertova, 2024: 23). UAS contest the immediate airspace above ground forces, as their small size, speed and low-altitude flight profile complicate detection and interception.

The integration of pervasive UAS into the battlespace has permanently added a vertical dimension to section-level tactical planning. It requires a task organisation which would have been considered an exceptional measure in pre-drone doctrine. Watling (2025a: 17) observes that Ukrainian sections now routinely divide their attention, evenly split between the ground and air — echoed by Zhluktenko (2026). This is not an adaptation to exceptional circumstances. The section commander who plans only for the ground plane now plans for less than half the battlespace.

Within the zone of contestation, advancing on enemy positions becomes prohibitively dangerous unless enemy drones are first suppressed, limiting opportunities for close combat. On quieter sections of the front, small unit infiltration, raids and ambushes can be effective. The importance of CUAS during offence is underscored by the observation that "if you don't use anti-drone measures [...] every mission will be a failure" (Ukrainian Lessons Learned SOF department, n.d.-b). This places a premium on route planning, dedicated CUAS groups, ER, EW, indirect-fire suppression and *hard-kill* countermeasures.

Contemporary Russian tactics illustrate the integration of these dynamics. Watling (2025a: 3–4) and Zabrodskyi (2023) describe an approach combining reconnaissance, suppressive fires and small unit infiltration. Forces conduct reconnaissance by fire, build combat power incrementally and exploit degraded enemy ISR and favourable conditions to advance. While effective, this approach exposes light infantry to significant attrition, as evidenced in 66th Separate Mechanised Brigade (2026).

For key objectives, dispersion alone is insufficient. Successful offensive action demands fleeting concentration through mobility and kill-chain disruption (Zabrodskyi et al., 2022: 62–63), supported by shaping actions which isolate the enemy via UAS suppression, EW and logistics strikes (Néron-Bancel & Garnier, 2024; Ukrainian Lessons Learned SOF department, n.d.-b) and (Watling, 2023b: ch. 9). Deception has regained prominence, understood through the see–think–do model, which requires it to induce specific enemy actions (or inaction) rather than mere belief.

Stepanenko (2025) assesses that AFRF FPV-OWADs are likely achieving limited Battlefield Air Interdiction (BAI), which raises the threat to Ground Lines of Communication (GLOC) and logistics. The vulnerability of Russian C2 when deprived of reliable connectivity — such as Starlink — illustrates this logic. Dave (2023), Hallatt (2023a), and Hallatt (2025) highlight the dangers of post-assault consolidation, as ISRD-enabled targeting rapidly brings accurate indirect and drone-delivered fires onto newly seized positions. Watling (2025a: 5–11) demonstrates that Ukrainian offensive action is no longer conceived as a rapid breach but as a phased contest for local conditions. The sector is first surveyed, then isolated from support and resupply, degraded, fixed and suppressed before assault forces close and consolidate.

Infantry can only close with the enemy as the terminal act of an attritional sequence. Assault is the exploitation of conditions created by prior shaping phases. Manoeuvre is thus generated not by speed or shock, but by systematically degrading the defender's capacity to observe, reinforce and respond across the grey zone. It aligns tempo with the control of the electromagnetic spectrum. Aggressive EW during suppression deliberately trades friendly RC UAS utility for enemy paralysis (Watling, 2025a), thereby heightening an existing premium on indirect fires identified by Interview Participant 04 (2026) and Interview Participant 08 (2026). For Western forces, these lessons underscore that effective combined-arms synchronisation under pervasive sensor coverage demands sequencing of effects across space, time and the electromagnetic spectrum — a *multi-domain operation*. Manoeuvre itself is a conditional phase achieved only once dominance in the middle battle is secured.

Defensively, ISR imposes further constraints. Administration outside of trenches is often impossible, requiring either interconnected trench systems or self-sustaining positions.

Robotics offer partial mitigation. Watling (2023b: 57) suggests that such systems can sustain dispersed forces and reduce exposure by enabling resupply without human movement. While subterranean systems offer protection, they can degrade effective defensive action. Firing ports, though essential for observation and fires, have become liabilities, with FPVDs targeting them directly and rendering even observation hazardous (Néron-Bancel & Garnier, 2024; Tyshchenko & Kiley, 2026). This indicates the requirement for CUAS barriers and unattended ground sensors to detect humans and electronic reconnaissance to detect RC drones (Zabrodskyi et al., 2022: 59).

Armed drones introduce an IED hazard that current light infantry capacity cannot resolve. Improvised munitions are readily adapted for armed drones — a practice documented on both sides of the Russo–Ukrainian War. In functional terms, armed UAS must be treated as IEDs (Óglaigh na hÉireann, 2022; Interview Participant 05, 2026). The improvised nature of firing systems compounds this hazard: Civ Division (2025c) and Interview Participant 01 (2026) report that Ukrainian systems generally include safe-arm switches, whereas Russian equivalents often do not. More sophisticated initiators — including doppler or microwave sensors and concealed switches — have also been reported, creating Victim-Operated IEDs (VOIEDs) (Terrogence Global, 2026a; Terrogence Global, 2026b). VOIEDs may be indistinguishable from inert or damaged UAS and can be initiated by a wide range of ordinary actions, including passive detection. This renders any enemy drone storage, workshop or launch site a potential Improvised Explosive Device Disposal (IEDD) task — one which cannot safely be cleared by non-specialists.

Successful CUAS does not remove this threat. Downed armed drones require render-safe at a scale which trained IEDD capacity cannot absorb[4]. This forces non-specialists to

[4]In personal communication with a Bundeswehr IEDD operator (2025), IEDD capacity was assessed as having been reduced by as much as two-thirds following a renewed emphasis on traditional combat engineering; no public source confirming this figure has been identified.

choose between tolerating the hazard and attempting precarious intervention (International Legionnaires | Ukraine, 2024b; Anonymous Veteran of the Russo-Ukrainian War, 2026: 2:58–5:44) — necessitating a decentralisation of technical expertise.

Armed drones therefore extend the IED threat vertically (Interview Participant 05, 2026), increasing the burden on small unit route discipline, clearance drills and hazard awareness. This reinstates the case for a dedicated assault pioneer[5] capability within the battalion support company. It raises the question of whether organic augmentation at section level is warranted (Lyle, 2023a: 28:14–30:47) and (Interview Participant 08, 2026).

UAS drive an evolution in the logic of survivability. For the United States Marine Corps (2016: 6) "to be detected is to be targeted is to be killed". Similarly, veterans such as Gifford (2025: 9:15–10:55) report that "the drone will not necessarily kill you, but it will lead to your death if it spots you". However, evidence from the Russo–Ukrainian War suggests a more nuanced reality, in which the kill-chain may stall after detection. The firer must weigh the value of the target against the cost of ammunition, the expected battlefield effect, competing fire priorities and the risk of revealing their own position and being targeted in return.

Survivability is therefore achieved less through protection or firepower than through becoming the *least interesting target* — echoed by Néron-Bancel and Garnier (2024: 53). The same logic applies electromagnetically: the communications problem may be less one of capability than of signature, where cheap, low-power, disposable radios that hide within civilian electromagnetic noise prove more survivable than exquisite systems that advertise their presence to an AI-enabled adversary (United States Marine Corps, 2016; Birchfield, 2026: 9, 18).

[5] Here, the role is taken to subsume both traditional infantry combat engineering and Assault IEDD; see Glossary.

For example, robotic CASEVAC may present a less attractive target than a group of stretcher bearers[6]. Similarly, visibly well-equipped formations attract greater attention than lower-signature troops moving along the same route. Examples include the Russian use of horses instead of vehicles to reduce target value (Axe, 2022). Interviewees report that foreign legionnaires attracted fire where conscripts on the same routes did not (Interview Participant 02, 2026; Interview Participant 04, 2026). Soldiers may even attempt deception, for example by *playing dead* (Magyar, 2026b). Infantry therefore adapt by maximising dispersion, camouflage, timing, unpredictability and signature discipline (Crawford, 2026). The survivability onion increasingly applies to dismounted infantry and not just vehicles (Hémez, 2017).

Dispersion preserves infantry survivability but imposes an *infantry ceiling*: the threshold at which dispersion precludes the concentration of mass required for breakthrough. Where the defender effectively integrates UxS into the kill-chain, the traditional 3:1 offensive heuristic becomes insufficient, as numerical mass increasingly generates targets rather than combat power.

The UAS threat renders CUAS an enduring light infantry requirement irrespective of operational context. Interview Participant 08 (2026) argues that the requirement for CUAS at sub-company level is contingent on operational context, increasing in static or positional warfare and diminishing under conditions of manoeuvre. Hvizda et al. (2025: vi, 22–23, 31) argue that the drone-enabled stagnation observed in Ukraine is not a permanent feature of warfare, but a consequence of neither side achieving air superiority. In their account, once NATO gains air control — assessed as the likely outcome of a NATO–Russia conflict given Western technological overmatch — persistent UAS-enabled surveillance

[6]Two interviewees reported being wounded by mortar fire while acting as stretcher bearers; see also UA Land Forces (2026b).

and strike become less decisive, enabling offensive ground manoeuvre. However, this argument applies only under specific conditions: where air superiority and manoeuvre are achievable. A US–China conflict over Taiwan may differ, producing a maritime and stand-off strike campaign rather than mobile combined-arms ground operations (Hvizda et al., 2025: 34–35). This is reinforced by Cancian et al. (2023), which predicts a positional, attritional defence centred on containing and reducing a beachhead. More fundamentally, even offensive operations are not necessarily characterised by manoeuvre at the tactical level, but by positional, fires-dominated combat (Fox, 2021; King, 2023). Taiwan therefore shows that the tactical problem is not confined to Ukraine: where an attacker must lodge, move and sustain forces under observation, UxS can turn exposure into a defensive advantage by improving the defender's RCP.

For light infantry — where UxS lethality is most acutely experienced — air superiority is tactically remote relative to their immediate exposure. UAS presence is immediate and continuous, requiring adaptation regardless of higher-level operational outcomes. This dynamic is reinforced in operations other than war, where air superiority cannot be assumed, rules of engagement constrain suppression and adversaries can employ UAS at low cost and low political risk (Watling & Bronk, 2024) — entrenching persistent exposure. While equipping units with layered CUAS assets to detect and engage UAS matters for force protection, hard- and soft-kill platoon-level CUAS self-defence is inherently constrained. Watling and Bronk (2024: 28) conclude that the battalion is the smallest unit able to support organic CUAS in a fully resourced fashion. The research nevertheless points to a requirement for CUAS effectors at all levels: while a complete organic capability may only be viable at battalion, the need for at least limited detection and self-defence reaches down to the section.

The underlying issue is contextual. In manoeuvre warfare, the most consequential UAS threat is typically ISRD-corrected fires. Once located, an infantry section is more likely to face suppression by indirect fires than FPVDs. This renders personal CUAS measures of limited operational significance. In this sense, the shotgun solution addresses the terminal symptom while the kill-chain closes around the shooter. However, the calculus differs in positional warfare, stabilisation/security/peacekeeping tasks and the defence of fixed installations. Here, drones may represent the primary threat and fires-based suppression of UAS teams is neither guaranteed, appropriate nor responsive. The targeting of a US base attributed to Kata'ib Hizbullah illustrates this dynamic: in such environments, attriting with inexpensive FPVDs becomes feasible precisely because the conditions which render personal CUAS marginal in manoeuvre warfare do not apply. This indicates the value of dispersed, decentralised CUAS effectors, as Hinote and Ryan (2024: 5) advocate.

Participant interviews and combat footage clearly elucidate the requirement for decentralisation. Interview Participant 04 (2026) is explicit that under drone ISR, sections must fragment into pairs and trios. EW-degraded communications is assumed rather than exceptional. As he notes, commanders must accept that they "don't have control" of their section for extended periods, relying instead on subordinate initiative when plans inevitably fracture under contact (Interview Participant 04, 2026).

US Department of the Army (2025: 61,196) identifies mitigating detection as essential, prescribing multiple axes, extended march intervals and concentration only sufficient to generate favourable force ratios. In Ukraine this manifests as packeted movement, staggered by intervals of 10–15 minutes — regrouping in hardened or subterranean rendezvous points. Dispersion does more than complicate centralised control: it precludes it. Hence, every soldier must know routes, timings and actions-on, even requiring privates to function as “critical thinkers” (Murray & Tuga, 2025: 1:04:00–1:32:00) — echoed by Leach (2025) and Interview Participant 01 (2026). This reflects a broader doctrinal shift toward mission command, comparable to the transition underpinning AirLand Battle, in which initiative at lower echelons became essential to combat effectiveness. Relinquishing control is psychologically demanding, yet unavoidable (Interview Participant 04, 2026). Interview Participant 09 (2026) identifies a tension whereby section commanders must occasionally order concentration to achieve tactical effect, thereby exposing troops to heightened drone risk.

Where surprise is unattainable, Néron-Bancel and Garnier (2024) argue that only tactical skill and mission command can compensate. This reinforces the value of competent platoon and section leadership. Crawford (2026) — a foreign advisor to AFUA — corroborates this directly: centralised control is increasingly impractical, making initiative at section and platoon level a tactical necessity. The implication is significant: pervasive ISR increases the demand for competent junior infantry leadership.

Murray and Tuga (2025: 5:20–6:26) attribute much of 3rd Assault Brigade’s battlefield performance to its western — rather than Soviet — procedures, specifically its empowerment of junior leaders. Murray and Tuga (2025: 1:04:52–1:32:00) describe the rapidly evolving nature of infantry TTPs as necessitating After Action Review (AAR) after each engagement — particularly for junior officers. Even apparently routine factors such as enemy unit rotation can produce markedly different enemy TTPs. Ego was identified as a major barrier to adaptation, while maturity was a key indicator of junior leader effectiveness. Interview Participant 04 (2026) described how overconfidence preceding a daytime platoon-assault on entrenched positions contributed to severe casualties, illustrating the tactical consequences of leadership hubris.

Platoon commanders retain a decisive role in pre-contact planning, coordination and fires integration. On contact, however, control fragments and execution devolves to dispersed fights (Interview Participant 04, 2026). UxS have increased the requirement for coordination, since the battalion is the lowest-echelon at which fires can be meaningfully planned (Watling & Reynolds, 2025b). Offensive action is therefore reconstituted as a sequencing problem in which the assault is the culmination of shaping and enabling actions. This logic is especially pronounced in urban terrain, where manoeuvre above company level is constrained (Interview Participant 08, 2026) and (Watling, 2023b: 144–147). Hence, decision-making authority — and the associated cognitive burden — is concentrated on section and platoon commanders. They must simultaneously prosecute the close fight and serve as information nodes for higher formations. Effective action increasingly depends on the initiative of corporals and lance corporals.

Chapter 1 concludes that ISRDs should be decentralised. While the OODA loop and kill-chain can be dramatically hastened by ISRDs, it can only be realised through communications — typically verbally or by radio. UAS teams are vulnerable to detection and fires and therefore require protection — optimised when sited within hardpoints or armoured vehicles. Hence possible advantages offered by ISRDs are vulnerable to the adversary's EW.

Axe's (2026a) account illustrates that Russia's drone-enabled, centrally controlled model — reliant on Starlink, Telegram and even live helmet-camera feeds to higher headquarters — proved brittle when connectivity collapsed. Ukraine exploited this silence in a manoeuvrist manner — conditions under which mission command, rather than centralised control, proves most robust.

Reports of poorly trained soldiers having their every battlefield movement dictated over unencrypted radio by a commander observing via UAS are common — depicted by FRONTLINE PBS (2025). Interview Participant 01 (2026) describes its utility in the real-time correction of gunfire and grenade throwing during a firefight.

International Legionnaires | Ukraine (2024a) and International Legionnaires | Ukraine (2025) document two assaults conducted by Chosen Company of the International Legion, at Pervomaisk, against the same defensive positions under similar conditions — supplemented by participant testimony (Hallatt, 2023b; Dave, 2024; Interview Participant 01, 2026; Interview Participant 04, 2026; Interview Participant 09, 2026). The first attack succeeded largely through a manoeuvrist approach. The second — executed with essentially the same scheme of manoeuvre — encountered an enemy which had adapted. Following the initial engagement, the defenders integrated ISR drones to correct indirect fires and control UAS strikes. It produced a decisive defensive action.

The episode illustrates a pattern described by practitioners: while the attacking unit repeated its earlier plan in part through "hubris" (Interview Participant 04, 2026) derived from previous success, the defending force appears to have conducted its own effective AAR and adaptation. The decisive variable was the defender's integration of ISR and strike drones into the kill-chain. The case therefore functions as a natural comparative vignette. It illustrates both how rapidly the drone-enabled battlespace punishes tactical repetition and how leadership ego or failure to adapt can magnify that vulnerability. The same failure to adapt operates at the institutional level, where bureaucratic inertia and organisational culture can entrench tactical obsolescence — a theme returned to in Chapter 3.

As warfare unfolds through continuous cycles of adaptation between adversaries, learning and innovation must occur at every level. This reinforces mission command's demand for decentralised decision-making and rapid bottom-up feedback (Ryan, 2022).

Future light-role sections must balance defender/assaulter specialisation, organic ISR/CUAS/UGS/assault pioneering integration, habitual versus organic specialists, and scalable command capacity. Interview Participant 01 (2026) recorded that drone pilots are exceptionally vulnerable when *under glass* (piloting under goggles). This thesis synthesises practitioner and expert testimony (Watling, 2023b; Interview Participant 01, 2026; Interview Participant 04, 2026; Interview Participant 08, 2026; Interview Participant 09, 2026) to propose two analytical constructs. Figure 2.1's notional baseline section comprises rifles, a machine gun and a *separate commander*. Figure 2.2's expanded section incorporates ISRD, CUAS, assault pioneering and security.

Figure 2.1 presents a notional nine-soldier baseline section: a commander (corporal), two detachment commanders (lance corporals) and two detachments — one configured around a grenadier and riflemen, the other around a machine gunner, loader and rifleman. The inclusion of lance corporals frees the commander's cognitive bandwidth and preserves simplicity. However, the commander must personally manage all communications and *systems*, creating a cognitive bottleneck (Interview Participant 08, 2026). This mirrors the British Phalanx Platoon Concept (PPC) nomenclature, where systems comprise added technology such as battlefield situational awareness (Lyle, 2023b). This structure remains reliant on higher-echelon CUAS which, during positional tasks, may constitute a *critical vulnerability*. Absent an organic ISRD, all reconnaissance must be performed by dismounted soldiers, exposing them to fire.

Organic capabilities at section level are therefore necessary. A dedicated ISRD/systems operator compresses the kill-chain and protects dismounts, while organic hard-kill CUAS is required to mitigate the latency and critical vulnerability inherent in reliance on battalion-level assets during positional fighting (Watling & Bronk, 2024: 28). Light infantry sections therefore require organic *hard-kill* CUAS to remain tactically viable. Meeting both requirements, while preserving C2 bandwidth and local security, demands additional junior leadership. The downed-drone-as-IED hazard alone justifies an organic assault pioneer — a requirement compounded by breaching, route proving and hasty fortification under persistent observation.

Accordingly, the smallest viable self-contained section comprises 14 personnel: a commander (corporal), three detachment commanders (lance corporals), two assault detachments of four (each led by a lance corporal and configured as shown in Figure 2.2) and a support detachment of five — integrating ISRD, CUAS, assault pioneer and a rifle — possibly a sharpshooter with a smart gunsight. This structure retains distinct command and manoeuvre elements, reduces the commander's cognitive burden, facilitates dispersed movement and action. It integrates organic ISR and CUAS at the lowest tactical level.

Mobility imposes a critical constraint across all models. A plausible configuration employs light 4×4 platforms to balance mobility, dispersion and survivability. In the context of light-role reconnaissance, Watling (2023b: 112–113) suggests a size comparable to a civilian pick-up carrying three soldiers per vehicle (for reserve capacity), reinforcing the need to distribute personnel across multiple platforms. Applying this criterion to six-seat 4×4 platforms, both the nine- and fourteen-soldier models require two vehicles, although multiple vehicles per section introduce non-trivial logistical and coordination burdens (Interview Participant 08, 2026).

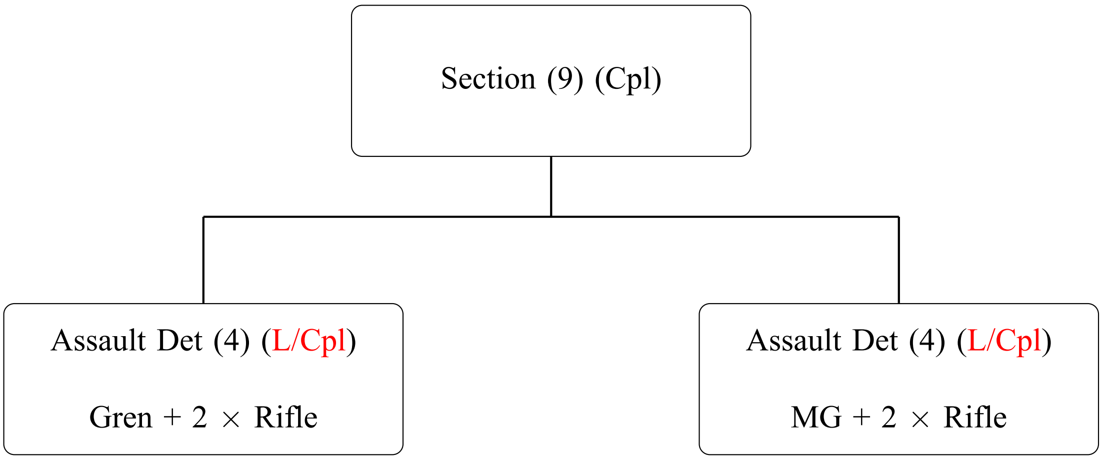

Figure 2.1: Section structure 1: Nine-soldier section with two assault detachments and separate commander

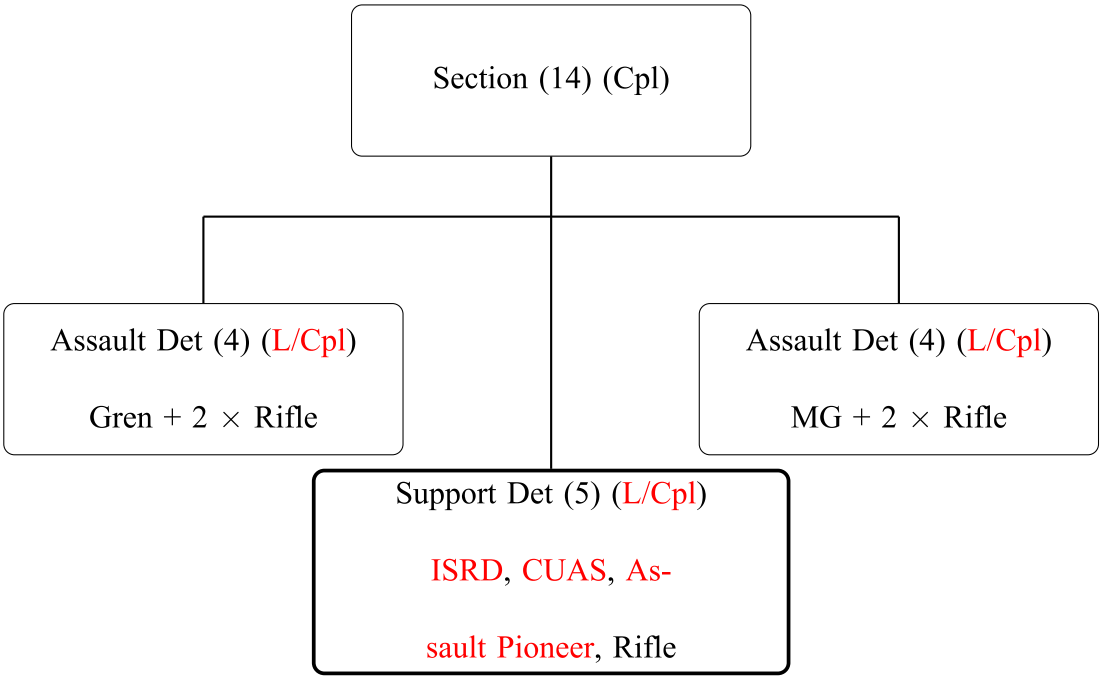

Figure 2.2: Section structure 2: Fourteen-soldier section with two assault detachments, support detachment and separate commander — the drone-survival minimum

Kahn (2024: 5:50–6:19) and Interview Participant 04 (2026) describe the significant cognitive burden placed on soldiers in battle. This burden is exacerbated by UAS, which demand simultaneous situational awareness of both the ground and aerial domains. In particular, drones enable ambushes, mines and IEDs to be deployed or repositioned rapidly, rendering previously cleared routes dangerous again. Readiness for contact through scanning arcs in the high-port position may result in tunnelled vision, diverting attention from equally significant threats from above and below.

Tensions arise where the risk of aerial observation in open terrain exceeds the threat posed by ground hazards. In such circumstances, soldiers may prioritise speed of movement, running across exposed ground rather than scanning or proving the absence of mines and IEDs — with rapid movement perceived as the only viable defence.

The psychological pressure is intensified by the constant auditory signature of drones. The high-pitched buzz alone causes soldiers to take cover instinctively, fostering a pervasive sense that nowhere is safe — a modern form of *drone shock* comparable to the tank shock of 1939–41 (Molloy, 2024; Néron-Bancel & Garnier, 2024). This effect may also serve as a deliberate tool of deception, inducing unnecessary reactive behaviour.

The cognitive burden is not confined to the dismounted soldier. Drone pilots themselves operate under substantial strain (Civ Division, 2026c: 3:20–5:37). This environment validates Néron-Bancel and Garnier's (2024: 50) observation that transparency can produce soldiers who see everything, but do not understand. Battlefield transparency imposes its own cognitive penalty. The same volume of sensor data which enables targeting can saturate the capacity to act on it. Perez (2026) and Néron-Bancel and Garnier's (2024: 24) concept of "infobesity" captures how this surplus of information paralyses decision-making at the analytical level. Under these conditions, persistent surveillance does not enhance control; it degrades it, producing *cognitive fratricide* and eroding the conditions required for mission command.

Combat training feedback loops have compressed dramatically. Practitioner experience now circulates laterally via Discord and Telegram, outpacing official channels by connecting soldiers facing identical tactical problems. “What is learned in combat this month often reshapes how soldiers train the next month” (Crawford, 2026).

Early UAS employment was ad hoc — soldiers from non-specialist roles flew drones alongside primary duties. Their battlefield dominance has since driven dedicated units like AFUA’s Magyar’s Birds and AFRF’s Rubicon Centre, exposing institutional lag behind tactical adaptation (Molloy, 2024).

This practitioner-driven dissemination is not exceptional. An analogous pattern was observed within the Irish Army on UNIFIL/UNDOF deployments, where soldiers adopted *Maps.me* when institutional GPS navigation was unavailable. Commercial tools filled critical gaps in speed and accessibility unmatched by defence systems.

Pervomaisk testimony demonstrated that continuous infantry TTP evolution occurs at the lowest tactical level and is enabled by mission command. ISRD and helmet FPV footage — including adversary material — enable sport-style ‘post-match’ analysis. Interview Participant 04 (2026) recorded how section-level learning is formalised through structured review processes, reinforcing the rapid adaptation cycle observed in combat. The similar importance of remaining abreast of technological evolution is exemplified by Murray and Tuga (2025: 1:02:00–1:04:52). Short, accessible lessons-learned material accelerate this cycle. Section-level influence now shapes organisational adaptation — Mingus and Steppe (2026): “soldiers and leaders at the squad level influence the enterprise more than most realize”.

Persistence and UxS have fundamentally reshaped light infantry warfare by making exposure, rather than contact, the primary source of risk. Battlefield transparency increases lethality asymmetrically, favouring the prepared defender and forcing dispersion, cognitive overload and rapid adaptation. Mission command is therefore not diminished but necessitated. The 14-soldier light infantry section — integrating organic ISR, CUAS, assault pioneering and additional junior leadership — emerges as the minimum viable structure for survivable and effective combat in a sensor-saturated battlespace.

These adaptations carry implications beyond the tactical level. A force designed around dispersion, democratised ISR and mission command is not simply more survivable — it imposes cost differently. How that cost-imposition translates into strategic defensive utility for a small, fiscally constrained state is the subject of Chapter 3.

3 IMPACT ON FORCE DESIGN AND DETERRENCE-BY-DENIAL

The defensive is a stronger form of war than the offensive

— *Clausewitz (2006: ch. V)*

The tactical changes identified in the previous chapters carry direct implications for organisational adaptation, force design and small state military utility. This chapter argues that low-cost UxS alter the relative combat power calculation in favour of a resource-constrained defender. They do so by lowering the cost of reconnaissance, precision strike and sustainment. The *strategic* consequence is that an attacker must plan for a defender able to impose disproportionate cost, delay and uncertainty without matching the attacker platform-for-platform. UxS change the material conditions from which forms of deterrence may be generated. They increase the defensive value of dispersed mission command empowered light infantry. Deterrence-by-denial is treated here as an effect of altered RCP and cost imposition, rather than a theory of deterrence. This matters for Ireland because conventional mass is politically and fiscally constrained. A small state which cannot replicate the power and scale of larger militaries may still complicate an adversary's planning calculus by making lodgement, movement and sustainment costly. The question is therefore not whether UxS replace existing forces, but whether they allow small states to extract a greater defensive cost-benefit.

Unusually, Ireland has no national security strategy — or policy per Gray (2018). The Defence Policy Review therefore functions as the state's de facto defence policy, assigning the Defence Forces the role of "military defence of the state" (Department of Defence and Defence Forces, 2024: sec. 3.2). Without a unifying political policy and supporting defence and military strategies, the concept of "military defence of the state" remains notional. In that gap, the Defence Forces' Strategic Military Planning Directive (2025) provides military planning guidance, but it cannot substitute for the absent political policy framework.

The absence of a clearly perceived external threat has reinforced an assumption that territorial defence is politically improbable, allowing force design decisions to prioritise continuity over adaptation and without articulating a *theory of victory*. Yet recent developments among Western states — most notably the United States' assertions regarding Greenland (United States Congress, 2025: h. 972) and Denmark's response (Statsministeriet, 2026) — demonstrate that scenarios once considered implausible can rapidly acquire strategic salience. The perception of threat is an unreliable indicator for future conflict. Failure to internalise the implications of UxS risks entrenching a force posture optimised for conditions which no longer exist. UxS offer an opportunity to optimise the expenditure of toil, blood and treasure for national deterrence. Rather than consciously treating UxS as niche capabilities, the Irish Defence Forces have, through institutional inertia and strategic-cultural lag, largely failed to engage with them. It results in a de facto position of exclusion by omission despite broader efforts to restore legacy capability.

Set against the Army Force Design Office's conception of a future force (Department of Defence and Defence Forces, 2024; Army Force Design Office, 2026), these considerations raise two interrelated questions: whether Ireland's political system will sustain significant investment in defence; and whether drone warfare alters the defensive value proposition of such equipment. The Irish Government accepted the Commission on the Defence Forces (CODF) Level of Ambition (LOA) 2 transformation, but funded half of the €3.4 billion requested *for 2025–2030* (Wall, 2025). Reports indicated that Ireland may procure armoured vehicles for circa €600 million (Cabirol, 2026; O'Connor, 2026). Estonia's cancellation of a €500 million armoured vehicle programme in favour of enhanced drone defence and CUAS (ERR News, 2026), reflects states heeding Zaluzhny's (2026) characterisation of the war; recognising that Cold War conceptions are no longer even 'the last war'. As Chapter 1 argued, parity on war's outbreak requires acknowledging drones as structural modifiers of warfare — reshaping platform utility, tactics and force design.

Technological acquisition does not in itself produce transformation. Krepinevich (1994) argued that adaptation, not invention alone, determines whether innovation becomes decisive — inter-war Britain possessed the tank concept but lacked the organisational structures to exploit it. Betts (1996) critiques militaries practising "conservative progressivism" — embracing new systems while preserving established hierarchies and habits. Rosen (1991: 7) adds that genuine innovation requires new concepts, not merely new equipment. L. E. Davis et al. (2014: 11)[7] extend this logic to UxS, arguing that they are not transformative in isolation and require doctrinal and tactical agility to retain relative advantage. The Russo–Ukrainian War reflects this logic: UxS generated relative advantage where forces reorganised around them, creating drone units, faster kill-chains and operator-to-manufacturer adaptation loops. Revising authority structures, accelerating kill-chains and actualising mission command are more difficult than procurement (Krepinevich, 1994; E. A. Cohen, 1996). Even where technological advantages appear evident, outcomes are determined by culture, bureaucracy, doctrine and the ability to integrate new systems into existing structures (Owens, 2002; Stimson Center, 2015). Acquiring UxS does not automatically maximise RCP since the military system must evolve to realise it. Without organisational adjustment, technological integration risks generating cost without increased

[7] Writing primarily about MALE systems, though the underlying logic applies.

battlefield effectiveness (Schaus & Johnson, 2018). As Krepinevich (1994) implied, invention without reorganisation is a dead end.

Low-cost UxS create disproportionate combat power through cost imposition, fundamentally altering the deterrence calculus for resource-constrained states. Ukraine's early resort to improvised uncrewed systems reflected necessity rather than novelty — a logic relevant to small states with constrained conventional capability (Kunertova, 2024: 8) and (Molloy, 2024: 10). Such systems offer asymmetric returns on investment (Interview Participant 08, 2026); for a small state such as Ireland, this supports a more credible deterrent posture under conditions of defensive resource scarcity. Molloy (2024: 10) notes the emergence of coordinated drone operations, combining systems with different capabilities during a single mission. This enables multi-functional battlefield effects.

E. Davis (2025) and Kunertova (2024: 8) contend that cheap drones challenge traditional principles of force concentration and cost-benefit — creating a "mass effect" of their own. Neuberger (2026) considers UAS' cost-benefit compared with tanks, GBAD and attack helicopters. Ireland's most recently procured batch of Block F Javelin ATGMs — with a four-kilometre range — cost circa €187,715. The reported 50–70 kilometre FPV-OWAD range and lower cost warrant a comparative cost-benefit analysis. Consider the FPV-OWAD of Table 3.1.

Component	Cost (€)
13” drone	€742
Fibre spool	€38.3 km^{-1}
1 kg explosive	€50 kg^{-1}
2 x Electric detonator	€100 det^{-1}
Total per drone	10 km = **€1,375** & 70 km = **€3,673**

Table 3.1: Estimated Fibre-Optic FPV-OWAD Cost (exclusive of VAT), per VGI-9 (2025), Defpoint (2026a), and Defpoint (2026b)

Tethered OWAD Range	Cost Ratio	Range Ratio (Javelin:FPV)
10 km	1:111	1:2.5
70 km	1:42	1:17.5

Notes: Rounded ratios were calculated from: the Javelin Block F unit cost €152,614 exclusive of VAT versus Table 3.1's tethered FPV-OWAD unit costs of €1,375 at 10 km and €3,673 at 70 kilometres; and Javelin's four-kilometre range.

Table 3.2: Cost and Range Ratios (exclusive of VAT): Javelin versus Fibre-Optic FPV-OWAD, per Interview Participant 07 (2026) and US Department of Defense (2020: 187)

The barrier to entry for sophisticated, precision-guided munitions at scale is collapsing. US procurement indicates FPV-OWADs are being standardised as attritable munitions: the Army cites PBAS air vehicles at $5,000 each, while a United States Marine Corps requirement targets up to 10,000 FPV-OWADs below $4,000 per unit (United States Army, 2025; US Government, 2025).

Ireland's procurement of 44 Javelin ATGMs[8] represents a major capital commitment to a high-cost precision munition. As shown in Table 3.3, the same budget delivers dramatically different levels of aggregate striking potential and operational redundancy when alternative OWAD solutions are considered. This heuristic comparison is not a procurement recommendation, but rather an illustration of the quantity-quality trade-offs inherent in light infantry defensive procurement. At the low-cost end, systems such as AFUA's *Molniya* OWAD compress the cost-per-effect to a degree that challenges the justification for high unit-cost ATGMs on mass-effects grounds alone. Beyond cost, ATGMs' thermal signature (Molloy, 2024: 22), operator exposure, weight and bulk constrain employability in two key ways: they limit ammunition carriage and complicate both dismounted and mounted transport. While FPVDs may struggle to achieve K-kills, they reliably achieve M-kills, immobilising the platform and fixing it for destruction — via CASDs or artillery.

Table 3.1 elucidates that RC are significantly cheaper than tethered UAS. They are customisable on the battlefield and quite resistant to ECM. Hence they are assessed as the control-system of choice for manoeuvre warfare — particularly for states which can achieve air supremacy and inhibit the enemy's jamming. States may prefer RC: the quickest, proven and simplest mechanism to deploy drones — nevertheless susceptible to countermeasures. Such cost-efficiency and flexibility are therefore contingent on the environment: Molloy (2024: 49) reports that during the 2024 battle for Bakhmut, "they had at least 15–17 EW system installations per kilometre", often reducing the effective range to 50 metres.

[8] The military requirement was identified as 88 ATGMs, per a paper staffed by the author as part of the approved business case — heightening the cost comparison.

ATGMs versus FPV-OWADs and Loitering-OWADs

Munition	**Unit Cost (incl. VAT)**	**Payload (kg)**	**44 ATGM budget (purchase power)**	**Javelin eq payload**	**Salvo qty (armd coy)**	**Salvo cost**
Javelin ATGM	€188 k	8	44	44	6	€1.126M
RC FPV-OWAD (low)	€836	1	9.9 k	1,234	100	€83.6 k
RC FPV-OWAD (mid)	€1,882	3	4.4 k	1,646	33	€62.7 k
RC FPV-OWAD (high)	€3,137	5	2.6 k	1,646	20	€62.7 k
RC Loitering OWAD	€31 k	5	263	165	20	€627 k
RC *Molniya* OWAD	€418	5	19.8 k	12,344	20	€8.4 k

Table 3.3: Comparative Anti-Armour Procurement Heuristic: ATGMs versus FPV-OWADs and Loitering-OWADs. See Table A.3 for methodology

Although UxS are sometimes described as a “game changer” (Interview Participant 01, 2026), the term is useful only if carefully bounded. Countermeasures exist: they can be jammed, spoofed, netted, shot down and mitigated by cover, camouflage and dispersion (Bronk, 2025; Interview Participant 04, 2026; Interview Participant 08, 2026; Interview Participant 10, 2026). UxS do not replace conventional human forces: they cannot seize or hold terrain, and their battlefield effects depend on integration with infantry, fires, logistics and C2 (Molloy, 2024; Bronk, 2025; Watling & Reynolds, 2025b; Interview Participant 08, 2026). Their significance lies in the cost-benefit problem they impose: mitigation consumes resources, drones place planning *constraints* and *restrictions* on tactical action and they can narrow the RCP gap between the resource-constrained defender and a conventionally superior attacker. UxS are game-changing because the barrier to entry for precision strike, rapid kill-chains and persistent ISR has collapsed. Like the machine gun, drones are best understood as an evolutionary system, ultimately integrated and normalised at progressively lower tactical echelons (John, 2025; Interview Participant 08, 2026; Oliviero & Halton, 2026). Normalisation indicates that militaries have internalised the increased penalty of exposure and adapted structures, a Critical Requirement within a force — able to sense, strike, protect, command and sustain.

The effectiveness of UAS is highly variable and dependent on context, countermeasures and target type. Reported probabilities of FPVDs reaching their target vary widely: from 70–75 percent in permissive conditions (Interview Participant 01, 2026) and (Bespalov, 2026: 18:15–18:50), to 20–40 percent or lower in contested environments (Watling & Reynolds, 2025b: 10) and (Bronk & Watling, 2024: 27). Survivability rates for ISR and strike drones vary similarly, reflecting differences in EW conditions, CUAS and operator proficiency (Zabrodskyi et al., 2022; Bronk & Watling, 2024). Tethered systems mitigate jamming but introduce new vulnerabilities, including physical severance and problems spooling, reportedly reaching failure rates as high as 50 percent (Ingvald, 2026: 23:50–24:05). For light infantry, the most consequential probability is survivability: once detected, survival rates may fall as low as 10 percent without effective CUAS (Zander et al., 2025: 13:17–17:50). Gifford (2025: 2:55–4:19) observed “that drone will hit you; if it spots you and has a good pilot, you will die”. Against armoured targets, multiple FPVDs are often required to achieve a kill, with reports suggesting that up to 20 strikes may be needed to defeat a main battle tank (Bronk & Watling, 2024; Watling, 2025a; Interview Participant 08, 2026) — though mobility kills are often sufficient.

The apparent low unit cost of drones can be misleading. As Molloy (2024: 25) notes, achieving a desired effect often requires multiple drones, while batteries, logistics and supply constraints impose additional costs. UAS effectiveness is therefore not a function of individual system performance, but of scale, integration and the ability to absorb attrition. While FPV-OWADs have attracted disproportionate popular attention, it is dropper drones which most directly serve light infantry needs. Their reusability, payload versatility and multi-role capacity — spanning section- and platoon-level CAS, forward logistics and route interdiction — make them the more consequential acquisition priority for resource-constrained light forces.

Tollast (2025) cautions against treating cheap drone-mass as an easily replicable solution. Ukraine’s model depends heavily on Chinese components and supply chains which many states could not readily reproduce in wartime. This limitation is reflected in recent US efforts to decouple their supply chains from China (US House of Representatives, 2025). Taiwan’s drone industry now produces over 120,000 units annually (The

Economist, 2026), indicating that Western-aligned manufacturing capacity is emerging despite supply chain constraints.

Bronk and Watling (2024) note that UxS deliver effects at a cost and scale unmatched by alternative capabilities — illustrated by Table 3.3. For Ireland, the relevant comparison is not with Europe's front-line states, which operationalise military deterrence through heavily funded conventional forces and societal mobilisation. Ireland is unlikely to replicate that model (Commission on the Defence Forces, 2022: 135–148). The question is therefore whether lower-cost mechanisms — including dispersed drone mass enabled by mission command — can improve RCP by making lodgement, movement and sustainment more costly for an attacker. Interview Participant 07 (2026) captures the point directly: "deterrence-by-denial just got a lot cheaper for a lot of countries, because you can just wreak so much havoc on any kind of a mass formation".

For island states, potential adversaries are logistically disadvantaged. Ground intervention requires maritime or airborne entry — which can be observed (and targeted) by UxS. Interview Participant 07 (2026) considers that the requirement to establish a lodgement for ground operations constitutes a Critical Vulnerability for conventional forces, which can be exploited by "non-standard forces". This logic is reinforced by Hvizda et al. (2025: 22), who note that democratised ISR and precision strike "could complicate offensive operations in a manner resembling the current stalemate in Ukraine". Reconnaissance can detect force-massing, degrade camouflage and concealment and frustrate manoeuvre. When combined with medium- or high-altitude long-endurance UAS and OSINT, this allows mobile defences to mass effects through economy of force and superior situational awareness.

Given Óglaigh na hÉireann's roots as *flying columns* — per O'Malley (1936), Barry (1949), and O'Malley (1982) — Ireland's weakness in conventional territorial defence may be partly offset by an irregular inheritance. Dispersed forces, local knowledge, concealment and mobility become more relevant when integrated with cheap precision strike. It is an opportunity to deliver credible and cost-effective denial through UAS air interdiction.

Iran's March 2026 *Operation Epic Fury* retaliation exemplified a similar logic. Large numbers of comparatively inexpensive drones saturated regional air defences and forced the conservation of high-cost interceptors (Hinote & Ryan, 2024: 4–5), while preserving ballistic missiles for higher-value targets (Bondar, 2026). Iran sustained a steady rhythm of drone attacks across multiple theatres, demonstrating how low-cost aerial systems can impose continuous economic, operational and psychological strain on a defender. UxS impose a cumulative cost by forcing a superior adversary to protect exposed movement and sustainment nodes over time. For a defender capable of trading space for time, attritable UGS and OWADs are a Critical Capability . Watling (2023b: 138–139) notes that even small salvos of loitering munitions can saturate point-defences. Predictable employment risks optimisation by the attacker, but iterative and irregular OWAD salvos can impose sustained pressure, force defensive resource allocation and generate cumulative attrition. This matters for Ireland because UxS do not need to be independently decisive to affect RCP. Repeated attacks on movement, sustainment and force protection can create material loss while also increasing political cost for casualty-averse democracies (Luttwak, 1996). In operations planning terms, UxS exploit a political vulnerability through attrition: the attacker's need to protect, sustain and justify the force becomes part of the defensive cost-imposition mechanism.

For small states unable to match an adversary conventionally, political bias toward short wars may create an exploitable vulnerability. R. S. Cohen and Gentile (2023) argue that repeated underestimation of war's duration reflects a structural bias toward rapid, low-cost victory in great-power decision-making — the very expectation a defender can target. UxS-enabled fires and organised irregular resistance can signal that intervention will become protracted, observable and materially costly. The effect is not to guarantee defeat of a superior attacker. It is to undermine the assumption that victory will be quick, cheap or politically containable. Deterrence therefore depends on capability, credibility and communication: Ireland must possess usable means to impose cost, field them credibly and signal that an adversary's movement would be exposed, delayed and punished.

The cost-imposition mechanism operates through the credible communication that victory — if achievable — will be difficult. That message becomes credible when adversary staff assess RCP and advise that the operation may generate politically unacceptable costs. Australian Government Department of Defence's (2024: 22) strategy of denial is instructive because it frames denial as a signal of credible military capacity, not simply a battlefield outcome: "designed to deter a potential adversary from taking actions [and] signalling a credible ability [to] deter attempts to coerce Australia through force". For Ireland, the implication is narrower but relevant: UxS-enabled cost imposition can support denial by raising the expected costs.

W. S. Murray (2008) argued that Taiwan should reject symmetric competition in favour of denial and cost-imposition through a distributed *porcupine strategy*. Paparo later articulated a comparable operational approach, describing the saturation of the Taiwan Strait with UxS to deny a rapid, low-cost victory (Rogin, 2024). More recently, Thomas et al. (2014) and Pettyjohn and Campbell (2026) shift the question from whether Taiwan can win conventionally to whether an adversary can absorb the chaos, casualties and uncertainty imposed by such a defence. The transferable lesson is that denial works by making rapid victory uncertain, not by matching the adversary symmetrically.

For Ireland, the equivalent mechanism is not a porcupine strategy, but dispersed light infantry enabled by mission command and integrated UxS. Such a force would be difficult to track en masse, particularly in increasingly urbanised terrain (Defence Science and Technology Laboratory, 2020: ii). It could deliver the protraction and attrition required to undermine great-power short-war assumptions. For a small state lacking conventional depth, this force represents both the mechanism of cost-imposition and the credible signal that territorial coercion would be neither quick nor cheap. Williams (2022) highlights early United States Marine Corps observations that drones, loitering munitions and light mobility platforms enable light infantry to impose disproportionate costs on heavier mechanised forces without maintaining comparable armour. Subsequent combat has borne this out.

This does not make UxS a substitute for infantry. Leach (2025: 6:05–12:55) argues that "we must stop minimising the importance of [well trained] infantry" and other conventional weapons. Drone teams require defence and without infantry, the adversary would exploit this vulnerability (Leach, 2025: 7:00–7:30). The introduction of guided ATGMs, for example, was once considered to signal the end of the main battle tank. In practice, however, massed artillery and manoeuvre displaced firing positions and degraded their battlefield effect (Interview Participant 08, 2026). UAS teams are subject to similar pressures: their effectiveness can be reduced through suppression, the disruption of their logistics or control nodes and manoeuvre — as evidenced by contemporary Russian practice (Watling, 2025a: 3).

Given that most contemporary UxS remain dependent on radio or satellite connectivity, lost communications risks converting a drone-centric defence into a liability, exposing insufficient infantry depth (Axe, 2026a) — exemplified by Russia's 2026 communications collapse (Axe, 2026b). Autonomous terminal guidance has been anecdotally reported in some systems. Fully autonomous swarms, however, have not yet been observed operationally. Their emergence, discussed by Civ Division (2025a), would significantly alter current assessments of UAS dependence on communications links.

As Ingvald and Alex (2025: 19:00–20:50) observe "you cannot win a war with just drones", a conclusion echoed by Molloy (2024: 25) and (Bronk, 2025). UxS remain enablers rather than substitutes for combined arms forces (Bronk, 2022; Watling & Reynolds, 2023; Cornish, 2024; Watling & Danylyuk, 2024; Bronk, 2025; Watling & Reynolds, 2025a; Watling & Reynolds, 2025b; Radec, 2026). For a small state designing deterrence around armed UAS, the implication is clear: resilience, not acquisition, is decisive. Redundancy is essential: resilient communications, decentralised stockpiles and mission-command empowered infantry. Without these, drone-enabled deterrence becomes a vulnerability.

Concurrently, Leach (2025) and Interview Participant 01 (2026) note[9], drone teams must have a high level of infantry skills to begin with — for infiltration and defence against adversary raids. In functional terms, this aligns them closely with light infantry roles; hence, drone teams *are a form of infantry* with an "outsize tactical value" (Hvizda et al., 2025: 26).

Effective employment of UAS depends on organisational concentration at higher echelons combined with controlled decentralisation at lower levels. The battlefield effectiveness of Magyar's Birds and Rubicon Centre indicates that armed drones are most effective

[9]Confer Civ Division (2024b) for a battlefield depiction.

when concentrated in dedicated units capable of mass and rapid repositioning. These observations indicate a requirement for brigade-level UAS units. Ukrainian practice reflects this trend: the national police's Joint Assault Brigade *Lyut* fields the *Kruk* battalion to disrupt enemy logistics (Lyut Joint Assault Brigade, National Police of Ukraine, 2026), while the Fury Brigade integrates armed and unarmed UAS and UGS (Militarnyi, 2026), demonstrating the institutionalisation of UxS at scale.

Force design is therefore a question of structure and institutional permission. The challenge is not whether drones require regulation, but whether they are regulated in a way that preserves scale, attritability and adaptation. Interview Participant 03 (2026) contends that oversight, training and accountability remain necessary. However, applying the administrative logic of conventional aviation to small battlefield drones — many of which are best considered consumables (Deveraux, 2022; Bronk & Watling, 2024; Griffiths & Ivas, 2025) and Zabrodskyi et al. (2022: 58) — risks suppressing the attributes on which their combat value depends (Zabrodskyi et al., 2022: 57–58) and (Watling, 2023a: 51:30–54:15). In peacetime, this creates a particular danger: institutional inertia can prevent adaptation until combat imposes it at greater cost.

At lower echelons, the offensive corollary is a mixed organic capability. FPV-OWADs offer range but are single-use, whereas CASDs are reusable but lack equivalent stand-off. A complementary mix is therefore indicated, with stabilised CASDs possessing sufficient utility to warrant decentralised use, such that infantry develop baseline proficiency (Leach, 2025: 24:00–25:00).

This organisational logic must be replicated for countermeasures. Watling and Bronk (2024: 2) argue for battalion-level CUAS/anti-aircraft systems and brigade-level CUAS units. The vulnerability of static positions to UAS attack — illustrated by Iranian strikes during Operation Epic Fury (Al Jazeera, 2026) — reinforces that reliance on higher-echelon protection is insufficient. Effective battalion design therefore requires an integrated approach combining concentrated UAS, decentralised employment and organic CUAS.

Below battalion level, a tension arises between concentration and decentralisation. UAS are relatively fragile, logistically demanding and operator-intensive (Interview Participant 01, 2026; Interview Participant 06, 2026), indicating that effective employment requires dedicated teams rather than dispersal across manoeuvre elements. This reinforces the need for protected operators and local security, further binding UAS employment to infantry. The increasing technical and cognitive demands of modern warfare also raise training thresholds, with estimates suggesting that minimum infantry training now exceeds twenty weeks (Watling, 2025b; Interview Participant 04, 2026).

UGS enable partial decentralisation of heavy firepower. Lyle (2023a: 40:00–42:00) suggests that weaponised UGS could deliver effects comparable to organic armoured support at company level, potentially redistributing capabilities previously held at brigade. However, the characteristics of UGS differ sharply from expendable UAS. Greater size and weight, drivetrain maintenance, battery dependency, armour integration, exposure to direct and indirect fires and their non-consumable nature render organic platoon-level inclusion impractical. Their management more closely resembles that of vehicle fleets than expendable UAS. UGS are therefore best assessed as battalion-level assets, held within the support and headquarters companies. Limited allocation to light rifle companies for logistics is prudent. This assessment is consistent with the thematic analysis and reflects a broader requirement for organic all-terrain lift capacity within infantry companies, a role which UGS in a logistics function may partially fulfil.

These developments also challenge the traditional structure of the Support Company. The continued separation of anti-armour capability into a dedicated platoon is increasingly questionable. A reorganisation toward UxS — including the establishment of a CASD platoon and the partial conversion of anti-armour and reconnaissance elements — better reflects contemporary battlefield dynamics. Earlier reform proposals, such as Egerton (2021), anticipated the integration of ISRDs but did not foresee its pervasive effect on lethality.

This logic has direct implications for the internal structure of the battalion, particularly at company and section levels. A more ambitious design inverts the traditional logic: the assault detachment is built around the FPVD — as the machine gun once became the organising weapon of the infantry section. Given FPVD's demonstrated value for reconnaissance, precision strike and protected lethality, and the acute vulnerability of pilots when flying, this reorientation is analytically justified.

Figure 3.1 presents the resulting fifteen-soldier section. The first assault detachment pairs an FPVD operator with three rifles under a lance corporal, providing protection and enabling concealed operation. The second retains the machine gun, preserving organic suppression. The support detachment integrates ISRD, CUAS, assault pioneer and rifle under a fourth lance corporal. It is more demanding in training and procurement than the nine-soldier model. It is also more flexible — and lethal.

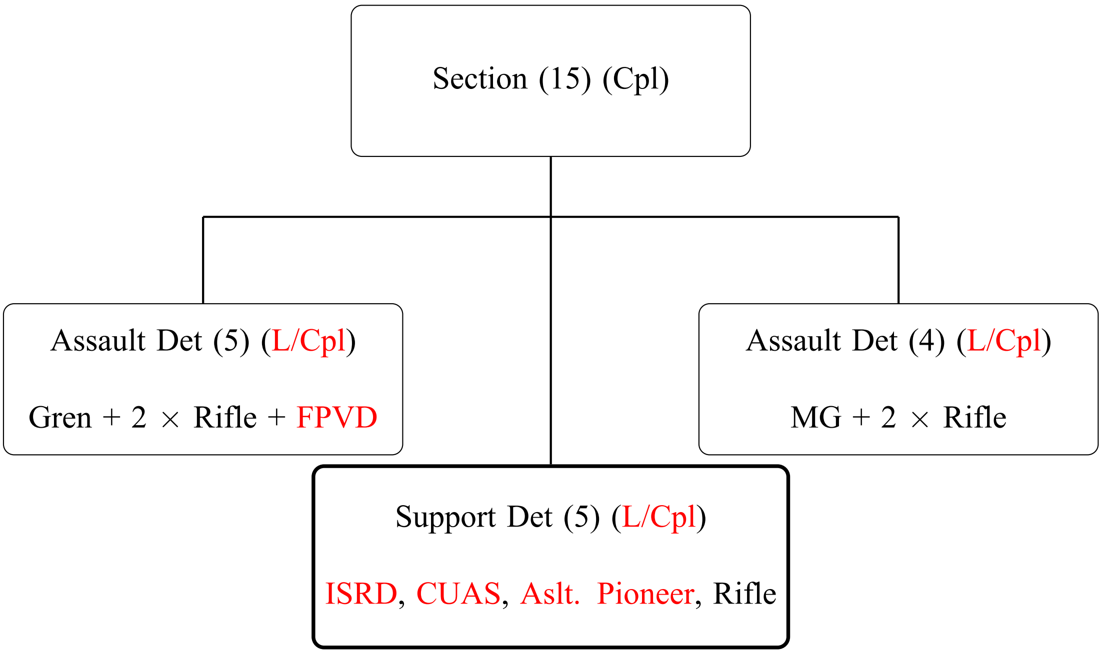

Figure 3.1: Section structure 3: Fifteen-soldier section with organic FPVD, support detachment and separate commander

This chapter has argued that UxS alter the cost structure of defensive RCP in ways directly relevant to Ireland's strategic position. The Javelin-to-FPV cost ratio — 1:42 to 1:111 depending on range — shows that mass precision effect is no longer the preserve of well-funded conventional forces. For a small island state whose irregular warfare heritage predates its conventional one, and whose political system has consistently declined to fund conventional mass at scale, UxS offer an alternative route to defensive utility.

The argument is not that drones replace conventional power. Rather, UxS can partially substitute for mass in selected defensive functions. They can make lodgement, movement, sustainment and consolidation more observable, targetable, delayed and costly. In doing so, they improve RCP by forcing an attacker to plan against a force able to impose cumulative cost without matching the attacker platform-for-platform.

That effect is only credible when UxS are integrated into a resilient force. Mission-command empowered light infantry remain the foundation. A drone-first posture built on fragile communications, foreign supply chains and insufficient infantry depth is therefore a vulnerability as much as a capability. Credible denial requires redundancy: resilient command, decentralised stockpiles, organic hard-kill CUAS, protected drone teams, secure supply and sustained political investment.

The central implication is therefore politico-strategic rather than technological. Ireland does not lack access to the basic technologies required to begin adapting. It lacks the strategic architecture, institutional urgency and political commitment required to convert those technologies into combat power. UxS do not solve Ireland's defence problem, but they clarify it: a small state can no longer justify weak territorial defence on the basis that credible combat power is financially unreachable.

CONCLUSION

Contemporary infantry warfare is increasingly organised around exposure: who is seen, when they are seen and how quickly observation becomes fire. Cheap ISR and the democratisation of precision strike mean that movement is no longer only a problem of terrain, but of timing and signature discipline.

Three analytical transformations stand out. First, movement as timed exposure treats movement as a risk-bearing act: every movement creates observability and must therefore be timed, concealed and protected. Second, persistent ISR imposes an infantry ceiling: dispersion preserves survivability but precludes the concentration of mass required for breakthrough, reshaping the conditions under which offensive action is possible. Third, conditional manoeuvre means that tactical success increasingly depends on sequencing ISR, EW and fires before assault.

These changes affect the RCP required for deterrence-by-denial. Low-cost UxS do not restore manoeuvre or replace conventional forces. They alter the cost structure of attrition, reconnaissance and precision strike in ways that favour a prepared defender. An attacker planning against a force which has integrated organic ISR, CUAS and armed UxS at the tactical level must account for disproportionate cost imposition at every phase of lodgement, movement and sustainment. A defender no longer needs to match the attacker platform-for-platform to impose cost.

For small states seeking credible denial, this shift is essential. Taiwan demonstrates that deterrence-by-denial can be generated by making rapid victory uncertain rather than by matching an adversary symmetrically. Ireland's problem is narrower but analogous: the Defence Forces are fiscally constrained, lack conventional mass and operate under an assumption that territorial defence is politically improbable. Mission-command empowered light infantry, integrated with low-cost precision UxS and resilient decentralised communications, represent a credible basis for cost imposition against a superior adversary. The limit of this posture is clear: drone-enabled defence built on fragile communications, foreign supply chains and insufficient infantry depth is a vulnerability as much as a capability. Credible military utility therefore demands redundancy, sovereign or allied supply and sustained political investment.

The implication for Ireland is uncomfortable. Treating territorial defence as improbable does not make it irrelevant; it allows force design to drift without a theory of victory. At section level, that drift is visible in the gap between the current nine-soldier section, the 14-soldier drone-survival minimum and a 15-soldier section built around FPVD. The nine-soldier section should therefore be treated as obsolete under pervasive UxS. Persistent ISR also makes lance corporal leadership a survivability and command requirement.

The practical recommendations follow directly. The Defence Forces should adopt the lance corporal rank; incorporate SLLS and movement as timed exposure into tactical SOPs, training syllabi and formal retraining; conduct structured experimentation with 14- and 15-soldier sections; reconsider the re-establishment of assault pioneering; and consider the licensing and regulatory arrangements needed to enable drone, CUAS and assault IEDD tactics. The wider implication is that the infantry section is becoming a micro-combined-arms unit, integrating ISR, air defence, engineering, IEDD and precision effects. Organisational bureaucracy, safety regulation and training systems must therefore enable adaptation, rather than obstruct it.

Future research should examine three areas: the empirical effects of drone shock on dismounted infantry performance and training resilience; the psychological impact of human–machine combat; and how Ireland could convert drone-enabled cost imposition into a coherent deterrence-by-denial strategy through capability planning, mobilisation, command resilience, stockpiling, supply chains and political signalling.

The question Ireland must answer is not whether uncrewed systems are affordable. It is whether the organisational adaptation required to exploit them is.

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Annex: Supplementary Evidence and Analytical Material

Thematic Analysis: Methodology and Coding Framework

The 178 primary sources were subjected to systematic bilateral thematic analysis using two coding approaches. Long-form sources, including documentaries, practitioner interviews and extended combat footage, were segmented into discrete tactical episodes and coded via spreadsheet across source, contextual and analytical fields, including source type, environment, contact type, episode type, troop disposition, survivability, sensing, movement, battle-drill behaviour, enemy and friendly drone presence, cue origin, unit size, initiative, outcome, leadership remarks and thesis implication. Short-form clips were organised directly into thematic categories reflecting the tactical phenomenon depicted: persistent ISR; ISR-corrected fires; battle damage assessment; FPV strikes against armour, vehicles, infantry and shelters; counter-drone action; drone-versus-drone engagement; close-air-support drone employment; UGS-enabled logistics and CASEVAC; armed UGS employment; human factors under drone threat; camouflage and signature management; drone-team survivability and munition preparation; movement under observation; and tactical adaptation under persistent surveillance.

Curated Examples of Tactical Behaviour

The examples below provide a curated evidential register of representative primary-source material.

- UAS targeting armoured vehicles: Hunter UA (2023), Sanders (2024), Sternenko (2024a), Sternenko (2024c), Ministry of Defense of Ukraine (2025), and UNITED24 Media (2025).
- Combined-arms destruction of armoured vehicles cued by and involving drones: Hunter UA (2023), Seveer of the 95th rifles (2023), Ministry of Defense of Ukraine (2024), Sternenko (2024b), and Sternenko (2024d).
- Flat Ukrainian steppe lending itself to optimal UxS performance: 72nd Separate Mechanised Brigade (2023) and elucidated by G. S. Jones et al. (2023).
- Daylight reconnaissance patrol: Tactical Group Athena and International Legion of DIU (2026) in a daylight reconnaissance patrol.
- Difficulty for UxS pilots to interpret the terrain through a 2D feed: Magyar (2026b).
- Detection being converted into almost immediate lethality: Azov Media (2024a).
- Use of UAS for real-time adjusting of fires, battle damage assessment and dynamic tactical decision making: Azov Media (2024d) and Wali (2025: 14:50–20:00).
- Point of view experience of dismounted infantry being targeted by an FPVD: Zander et al. (2025).
- Being targeted by a CASD: Trofimova (2024: 1:53–1:58).

- Vulnerability of entrances to defensive positions to drones and infantry assault: Azov Media (2025b), 225th Separate Assault Regiment (2026), Russian Ministry of Defence (2026b), Third Army Corps (2026a), and UA Land Forces (2026a).
- Primary source testimony of defending against FPVDs: Tyshchenko and Kiley (2026).
- Exceptional vulnerability of light forces to UAS: Azov Media (2024b) and Azov Media (2024c).
- FPV drones increasing infantry lethality at night: Zander et al. (2025).
- Waiter FPV drone: National Guard of Russia (2025).
- Use of UGS to mitigate human exposure to threats: Third Army Corps (2025b).
- UGS' greater difficulty than UAS in traversing terrain: 63rd Brigade (2026).
- Unarmoured CASEVAC UGS variant: Third Army Corps (2026b).
- UGS as disaggregated support by fire platforms: Third Army Corps (2025d).
- UGS as machine gun platforms during the assault: 22nd Mechanised Battalion, 3rd Assault Brigade (2026).
- UGS inducing surrender of the enemy: DevDroid LLC (2026) and Ministry of Defense of Ukraine (2026).
- Russian soldier evading five FPV-OWADs, while being observed by an ISRD: Stewart (2025: 9:36–10:38).
- Robotics mitigating the vulnerability of light forces to encirclement and bypass: Kirichenko (2026).

- Light infantry assaults remain possible in the most contested areas of the battlefield: Revanche Tactical Group (2026).
- A detachment-level assault: Blue (2026).
- Aggressive and decisive movement through the grey zone: International Legionnaires | Ukraine (2024c).
- Commanders perched cybernetically: Civ Division (2025d: 6:33–8:43) and FRONTLINE PBS (2025).
- UAS used to correct grenade throwing during the battle: International Legionnaires | Ukraine (2024c).
- UxS as IEDs: Magyar (2022) and Civ Division (2024a), Civ Division (2026a), Civ Division (2026f: 7:30–7:37) and discussed with Interview Participant 05 (2026).
- Defence of a moving vehicle from an FPV-OWAD by a shotgun: Civ Division (2025b: 0:16–0:50) illustrates.
- Live reflection on one's own battle via UAS footage: (Hallatt, 2023b; Dave, 2024).
- Short, accessible lessons-learned material: Ukrainian Lessons Learned SOF department (n.d.-a).
- Thermal camouflage held above the body in a manner analogous to the scrim umbrella: Civ Division (2025b: 0:00–0:35).
- Vulnerability of defensive firing ports to FPV-OWADs: Azov Media (2025b) and Russian Ministry of Defence (2026b).

Translations of Non-English Source Author Names

These tables provide the original non-English source author names and the English forms used for in-text citation and bibliography entries throughout the thesis.

Original Author Name	English Author Name Used in Thesis
Russian sources	
83 Уссурийская	83rd Ussuriysk Air Assault Brigade
Министерство обороны Российской Федерации	Russian Ministry of Defence
Родная 98-я ВДД	98th Guards Airborne Division
Росгвардия	National Guard of Russia

Table A.1: Translations of Russian Source Author Names Used in the Thesis

Table A.2: Translations of Ukrainian Source Author Names Used in the Thesis

Original Author Name	English Author Name Used in Thesis
Ukrainian sources	
2-й механізований батальйон 3 ОШБр	2nd Mechanised Battalion, 3rd Assault Brigade
225 ОШП	225th Separate Assault Regiment
47 окрема механізована бригада “Магура”	47th Separate Mechanised Brigade “Magura”
63 бригада | Сталеві Леви	63rd Brigade | Steel Lions
66 ОМБр ім. князя Мстислава Хороброго	66th Separate Mechanised Brigade named after Prince Mstyslav the Brave
72 ОМБр ім. Чорних Запорожців	72nd Separate Mechanised Brigade named after the Black Zaporozhians
Азов Media | Azov Media	Azov Media
Військово-Морські Сили ЗС України | UA Navy	Ukrainian Navy
Держприкордонслужба	State Border Guard Service of Ukraine
Kraken | ПБС 3 армійського корпусу	Kraken | Special Purpose Battalion, 3rd Army Corps
Мілітарний	Militarnyi
Об’єднана штурмова бригада НПУ “Лють”	Lyut Joint Assault Brigade, National Police of Ukraine
Птахи люті	Birds of Fury
Служба безпеки України	Security Service of Ukraine

continued on next page

Original Author Name	English Author Name Used in Thesis
Сухопутні війська | UA Land Forces	Ukrainian Land Forces
Тактична група “Реванш”	Revanche Tactical Group
Третій армійський корпус	Third Army Corps

Primary Source Interviews

Primary source interviews conducted for this thesis are anonymised as Participants 01–10. Publicly available primary-source interviews are cited using the name, callsign, handle or alias under which the source is publicly identified.

Participant	Background	Date
P01	Russo–Ukrainian War veteran, private	23 Jan 2026
P02	War correspondent; analyst	28 Jan 2026
P03	Irish military general officer	5 Feb 2026
P04	Russo–Ukrainian War veteran, junior NCO	6 Feb 2026
P05	Senior EU civil servant	6 Feb 2026
P06	Russo–Ukrainian War veteran, private	6 Feb 2026
P07	USSOCOM officer	23 Feb 2026
P08	Defence researcher	12 Mar 2026
P09	Russo–Ukrainian War veteran, junior NCO	4 Apr 2026
P10	Defence academic	4 Apr 2026

ATGMs versus FPV-OWADs and Loitering-OWADs

Two comparative criteria are applied: the “major defense equipment (MDE)”-only benchmark for 44 Javelin ATGMs (€8,259,450 incl. 23 percent VAT) and the 100-kilogram aggregate-payload equivalence.

The cited Foreign Military Sales (FMS) case was $8.7 million total, comprising $7.9 million in MDE for 44 FGM-148 Javelin missiles and $0.8 million in non-MDE support items per Federal Register (2026). As the comparison is expendable-to-expendable, only the $7.9 million MDE missile cost is used: $7,900,000 × €0.85 = €6,715,000 exclusive of VAT, or €8,259,450 inclusive of 23 percent VAT; €187,715 per missile inclusive of VAT (€152,614 exclusive). Non-MDE support items are excluded from the comparator.

The budget-equivalent Javelin effect and salvo quantity columns are independent. The former expresses aggregate striking potential purchasable within the 44 ATGM budget (€8,259,450 inclusive of 23 percent VAT). The latter expresses the number and cost of effectors required to defeat an armoured company. For OWAD variants this is calculated against the 100 kilogram shaped-charge warhead requirement assessed by Bronk and Watling (2024: 26–28); for Javelin, it is calculated as six missiles, assuming 100 percent kill probability against the 40–60 percent immobilisation threshold.

FPV-OWADs are assessed at one, three and five kilogram warhead payloads for the low-, mid- and high unit-cost variants respectively — though all rotary-wing FPVs can carry up to 5 kg, reduced payload preserves manoeuvrability and effective range at lower cost points.

The Javelin-equivalent payload column normalises each variant’s aggregate purchasable warhead mass to the Javelin’s 8 kg warhead: it is the number of units the 44 ATGM budget purchases, multiplied by that variant’s payload, divided by 8. It expresses each option’s total deliverable warhead mass in Javelin-warhead units.

Unit costs: FPV-OWAD — $800, $1,800 and $3,000 — and Loitering OWAD — $30,000 — per Bronk and Watling (2024: 27–28); *Molniya* — $400 — per Grobovy (2026: 1:15–1:21); Javelin

— \$7.9 million MDE for 44 missiles — per Federal Register (2026). Exchange rate \$1 = €0.85 applied throughout; all euro figures in Table A.3 include 23 percent VAT.

Munition	**Unit Cost (incl. VAT)**	**Payload (kg)**	**44 ATGM budget (purchase power)**	**Javelin eq payload**	**Salvo qty (armd coy)**	**Salvo cost**
Javelin ATGM	€188 k	8	44	44	6	€1.126M
RC FPV-OWAD (low)	€836	1	9.9 k	1,234	100	€83.6 k
RC FPV-OWAD (mid)	€1,882	3	4.4 k	1,646	33	€62.7 k
RC FPV-OWAD (high)	€3,137	5	2.6 k	1,646	20	€62.7 k
RC Loitering OWAD	€31 k	5	263	165	20	€627 k
RC *Molniya* OWAD	€418	5	19.8 k	12,344	20	€8.4 k

Table A.3: Comparative Anti-Armour Procurement Heuristic: ATGMs versus FPV-OWADs and Loitering-OWADs

Glossary of Terms

Assault IEDD The light-infantry application of IEDD to enable assaults and maintain momentum, conducting rapid render-safe procedures against access-denial (impedance) devices. During the GWoT it was largely regarded as a special-forces enabler. This thesis treats it as a conventional light-infantry requirement and proposes that it fall within a re-established assault pioneering discipline.

Assault Pioneering The light-infantry application of combat engineering, organic to an infantry battalion, providing light field-engineering support — defensive positions, base construction, water treatment, mine laying, breaching and detection — in direct support of the manoeuvre force (BFBS Forces News, 2020). Throughout this thesis, the role is taken to include Assault IEDD.

Attritional Warfare The [...] theory of attrition states that [belligerent] A seeks to change the relative balance of power in relation to [belligerent] B through the use of proximal and temporal firepower overmatch in consideration of both A's and B's battlefield disposition, for sustained periods of time, sufficient to gain its (A's) military objective (Fox, 2025: 39).

Battle Drills Also known as tactical standard operating procedures (TSOPs). They are predefined, rehearsed and well-understood actions or sequences executed in response to specific battlefield events. Their purpose is to reduce friction and enable rapid tactical action, minimising the need for further orders. They are triggered by the event itself rather than by a command.

Battlefield Air Interdiction (BAI) United States Air Force (1984: 3-3–3-4)[10] states that "objectives are to delay, disrupt, divert, or destroy an enemy's military potential before it can be

[10] See also McCaffrey (2001).

brought to bear effectively against friendly forces [;] Air interdiction attacks against targets which are in a position to have a near-term effect on friendly land forces are referred to as battlefield air interdiction".

Centre of Gravity A CoG is the primary source of power that provides an actor its strength, freedom of action, or will to fight. It is always a physical entity. At the political strategic level, moral-strength as well as physical-strength CoGs exist. At lower levels of command, only physical-strength CoGs normally exist. (UK Ministry of Defence, 2019: B-1).

Close Air Support Drones (CASD) Dropper drones used for close air support of frontline units. Also referred to as a "bomber drone".

Critical Capability What the CoG can do — its primary abilities — in relation to achieving the actor's objectives at the given level in the context of a given environment. The critical capability concept is useful to identify and validate CoGs, as it expresses how an actor can use a particular strength (the CoG candidate) to achieve the actor's objectives at the analyzed level of command (UK Ministry of Defence, 2019: B-7).

Critical Requirement Are specific conditions, resources, and/or means that are essential for a CoG to perform its critical capabilities (UK Ministry of Defence, 2019: B-7).

Critical Vulnerability Are those critical requirements, or components thereof, that are deficient, missing, or vulnerable to influence in a way that will contribute to a CoG failing to perform one or more of its critical capabilities. The lesser the risk and cost, the better (UK Ministry of Defence, 2019: B-7).

Doctrine "Fundamental principles by which the military forces guide their actions in support of objectives. It is authoritative but requires judgement in application" (Supreme Headquarters Allied Powers Europe, 2023: K-3).

Dropper Drone Rotary-wing UAS (generally with a frame-size greater than 10 $''$) used to drop explosive payloads — such as hand grenades, point-detonating ordnance and IEDs. Unlike FPV-OWADs, dropper drones are not consumable: they can be recovered, rearmed and re-tasked between sorties. Depending on control architecture, they may employ either manual first-person control (CAS-FPVD) or assisted-flight stabilisation, as with typical ISRDs. A dropper drone used for the close air support of frontline units is termed a Close Air Support Drone (CASD).

Electronic Reconnaissance (ER) The detection, location, identification and evaluation of the adversary's EM radiations (Joint Chiefs of Staff, 2012: I–9).

Electronic Warfare (EW) Military action involving the use of EM energy and directed energy to control the EMS or to attack the enemy. EW consists of three divisions: electronic attack, electronic protection and electronic warfare support (Joint Chiefs of Staff, 2012: viii).

First-Person View Drone (FPVD) A category of UAS which is flown through goggles in the pilot's view. They have become synonymous with kamikaze and tactical OWA drones. However, FPVs can also function as CASDs — with revolving drums utilised to produce 'bombing runs'. For this reason, this thesis distinguishes between FPVDs broadly, FPVDs used as strike munitions (FPV-OWADs) and those providing close-air support (FPV-CASDs).

Grey Zone Refers to the 20–30 km area forward of the FLOT that has become increasingly dangerous to traverse due to persistent ISR and the prevalence of indirect and uncrewed fires (Interview Participant 01, 2026; Texty.org.ua, n.d.). This usage is distinct from the term as employed in international relations. The "kill zone" is a subset of the grey zone. It may also be described as the *zone of contestation* or *contested zone* (Watling, 2025a: 5–6) (Watling, 2023a: 17:21–22:37).

High-Payoff Target "A target of significance and value to an adversary, the destruction, damage or neutralisation of which may lead to a disproportionate advantage to friendly forces" (Supreme Headquarters Allied Powers Europe, 2023: K-4).

Hunter Killer Drone An FPVD flown into the battlespace to search actively for enemy activity and engage suitable targets of opportunity — as distinct from the pre-positioned waiter. Such drones typically use an improvised point-detonating fuze and/or RC initiation; a typical point-detonating fuze comprises an open switch of two bare wire loops on the drone's nose which close on impact.

Improvised Explosive Device (IED) "A device placed or fabricated in an improvised manner incorporating destructive, lethal, noxious, pyrotechnic or incendiary chemicals, designed to destroy, incapacitate, harass or distract" (Óglaigh na hÉireann, 2022: 50).

Intelligence, Surveillance and Reconnaissance Drone (ISRD) Fixed-wing or rotary-wing UAS which operate below the coordinating altitude. For the purposes of this thesis, rotary-wing UAS shall be inferred unless otherwise stated. While FPVs *can* be employed, their continuous manual-piloting complicates their ISR applicability — they cannot hover. Hence, assisted-flight UAS are the norm — such as Parrot Anafi and DJI Mavic 3.

Kill Zone This is the section of the "grey zone" closest to the FLOT, where — due to the porosity and dispersion of the defensive dispositions — opposing positions are intermingled (Texty.org.ua, n.d.). Its width varies — reportedly being as wide as 10 km in Ukraine. This will vary by conflict and is likely to be a function of the terrain.

Kill-Chains "Dynamic targeting is executed using the dynamic process of F2T2EA. [...] Targets of opportunity have been the traditional focus of dynamic targeting because decisions on whether and how to engage must be made quickly. [...] The steps of dynamic targeting may be accomplished iteratively and in parallel. The find, fix, track, and assess steps tend to be intelligence, surveillance, and reconnaissance intensive, while the target and engage steps are typically labor-, force-, and decision making-intensive.” (Department of Defense, 2018: II–23).

Long Range Reconnaissance (LRR) “Extended-range reconnaissance, surveillance, target acquisition, and force protection over wider operational areas” (Center for Army Lessons Learned, 2025: 28).

Medium Range Reconnaissance (MRR) “Reconnaissance, surveillance, and target acquisition support for company- and battalion-level operations” (Center for Army Lessons Learned, 2025: 28).

Named Area of Interest (NAI) A location identified during Intelligence Preparation of the Battlespace (IPB), derived from terrain analysis and assessment of the enemy’s doctrine and likely courses of action. NAIs are areas assessed as potentially significant for observation in order to confirm or deny enemy activity. The number of NAIs will typically exceed available collection assets; consequently, not all NAIs can be observed at any given time.

One Way Attack Drone (OWAD) Long-range and slow-moving UAS intended to attack the enemy in depth and to supplement conventional cruise and ballistic missile fires (Bronk, 2025). Watling (2025a) distinguishes them from FPVDs in that they are not actively piloted.

Positional Warfare “Combined arms manoeuvre [...] whereby movement and fires undermine the enemy’s ability to hold key terrain” (Watling, 2025a: 1).

Relative Combat Power (RCP) It is a heuristic to assess military strength, analogous to the Lanchester Force Equations. It is assessed through a “matrix providing detailed information on the relative combat power using factors across the physical, moral and conceptual frameworks” (Óglaigh na hÉireann, 2016a: 16) and (Óglaigh na hÉireann, 2016b: 256), where common units and platforms are mathematically equated.

Short Range Reconnaissance (SRR) "Very short-range to short-range reconnaissance, direct situational awareness, and security for dismounted troops, squads, platoons, and companies" (Center for Army Lessons Learned, 2025).

Small Uncrewed Aerial System (sUAS) A UAS with a maximum take-off weight below a defined threshold — typically $25\ kg$ — employed for tactical reconnaissance, surveillance and target acquisition at sub-company level.

Target Area of Interest (TAI) A Named Area of Interest (NAI) where fires must be delivered in accordance with the targeting plan. TAIs are linked to decision points and enable the rapid engagement of identified enemy forces.

Transparency (of the Battlefield) "The ability to acquire and exploit geolocated, near-real-time awareness of a given operational environment thanks to a connectivity architecture linking networks of heterogeneous and redundant sensors, mass data processing systems, and effectors" (Néron-Bancel & Garnier, 2024: 3).

UAS Classes There is no agreed-upon standard for classifying UAS. Characteristics such as mass, range and ceiling are often used. This thesis concurs with Bronk and Watling (2024: 5) and considers five classes: "Situational awareness UAVs optimised for tactical reconnaissance; tactical strike UAVs; ISR UAVs able to penetrate into operational depth; operational strike UAVs; and platform-launched effects designed specifically to synchronise with and enable other weapons systems".

Victim-Operated IEDs (VOIEDs) An IED initiated by the actions of the victim, typically through a pressure plate, tripwire or similar mechanism.

Waiter Drones (waiters) The pre-deployment of a monitored FPVD to a location in anticipation of enemy activity. Modified with skids so that it can land and take off, allowing it to lie in wait, the drone is activated and attacks when a target presents; when none presents, it rests passively on the ground awaiting RC instructions. Commonly referred to as a “waiter”, “sleeper” or “ambush” drone.

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